

VERTICALLY-INTEGRATED DUAL-CONTINUUM
MODELS FOR CO₂ INJECTION IN FRACTURED
GEOLOGICAL FORMATIONS

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Abstract

Injection of CO₂ into a geological formation leads to a two-phase flow system, with supercritical CO₂ and brine being the two fluid phases. Two types of modeling approaches, three-dimensional (3D) models and vertical-equilibrium (VE) models, are traditionally used to study the two-phase system in an unfractured formation. 3D models solve governing equations in three dimensions and are applicable to generic geological formations. VE models assume rapid and complete buoyant segregation of the two phases, resulting in vertical pressure equilibrium and allowing relatively simple integration of the governing equations in the vertical dimension. This reduction in dimensionality makes VE models computationally less costly than 3D models, but restricts the application of VE model to relatively permeable formations.

A fractured geological formation is often modeled as an overlap of two continuous domains, one representing the fractures and the other representing the rock matrix. Both domains are usually treated as porous media continua and, as such, can be modeled by either a VE or a 3D formulation. In this thesis, a new model that combines both modeling approaches is developed. It applies the VE formulation to the fractures and the 3D model to the rock matrix. This model is compared to the Full-3D model, which uses 3D models for both the fractures and the rock matrix. The new model that couples a 3D model with a VE model, referred to as VE-3D, poses new challenges, which are analyzed and solved in this thesis. Simulation results show that for the test case used in this thesis, the VE-3D

model compares well with the Full-3D model in terms of fluid saturation, pressure buildup, and CO₂ mass partitioning in the fractured formation, but the VE-3D model incurs significantly lower computational cost.

Dual-porosity models, which ignore large-scale flow in the matrix, were also implemented by the author and are concisely discussed in this thesis. Coupled with the more comprehensive (dual-permeability) VE-3D and Full-3D models, this set of modeling options provide a range of effective computational tools to simulate CO₂ injection, estimate CO₂ storage capacity, and assess the risk of CO₂ storage in fractured subsurface formations.

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Chapter 1

Introduction

Carbon Capture and Sequestration (CCS) is the process that captures carbon dioxide (CO₂) from large point sources (e.g., fossil fuel power plants and cement factories), transports compressed CO₂ via ships or pipelines, and injects CO₂ into deep underground rock formations. It is the only currently available technology that allows hydrocarbon combustion to continue to power the economy while mitigating carbon emissions (*Celia et al.*, 2015) and is widely viewed as a bridge to a low-carbon future. *International Energy Agency* (2016) envisioned CCS to contribute 14% of annual carbon emission reductions in 2050, following the path toward the 2-degree Celsius climate goal.

Geological formations that are suitable for CO₂ storage are high-permeability formations more than 800 meters beneath the surface (*IPCC*, 2005) and overlain by a low-permeability formation (caprock) that impedes the upward motion of CO₂ caused by buoyancy. CO₂ can be injected and stored in different types of geological formations,

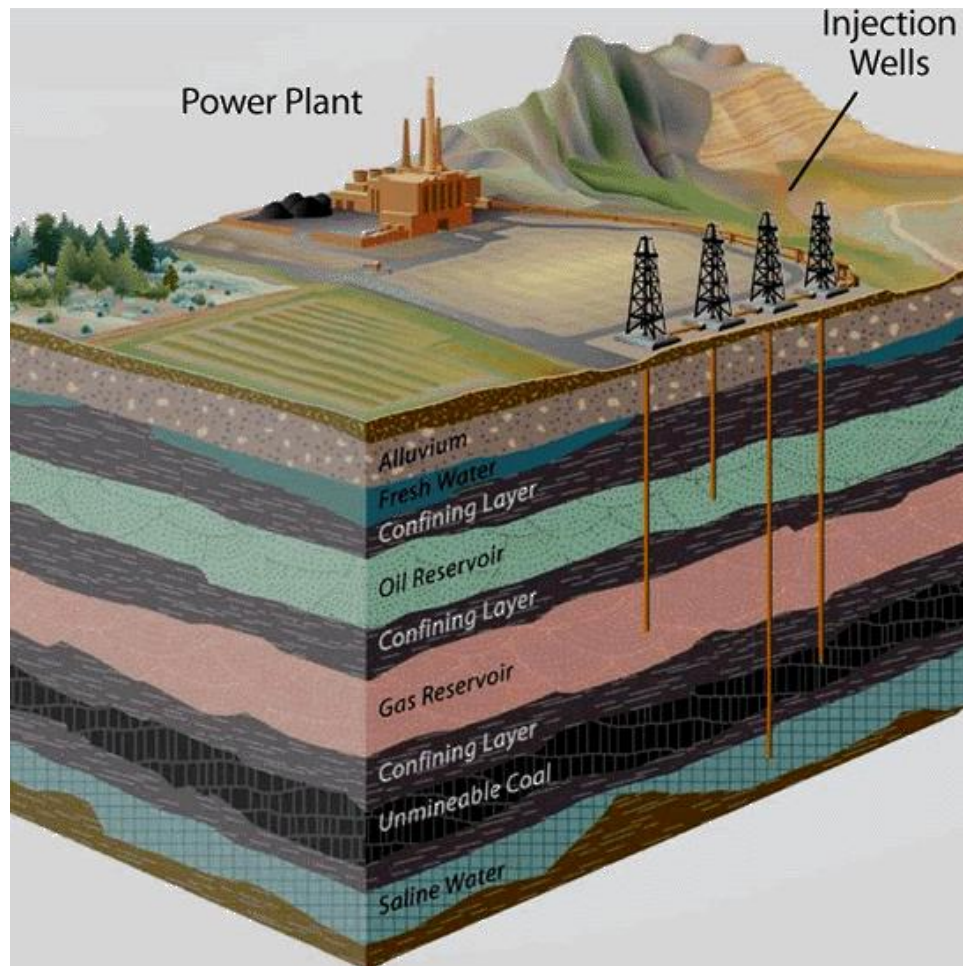


Figure 1.1 Picture showing the injection of CO₂ into different types of geological formations at different depths (modified from *Carbon Science & Engineering Research Group, University of Utah*, image produced by *Genevieve Young*).

such as depleted oil and gas reservoirs, deep saline aquifers, and unmineable coal seams (Leung *et al.*, 2014). This thesis focuses on CO₂ sequestration in deep saline aquifers because they are estimated to have significantly larger storage capacity than oil and gas reservoirs and are much more ubiquitous globally than hydrocarbon reservoirs (IPCC, 2005; Celia *et al.*, 2015). Szulczewski *et al.* (2012) showed that there is enough capacity in deep saline aquifers in the United States to store at least a century's worth of the current CO₂ emissions from all coal-fired power plants in the U.S.

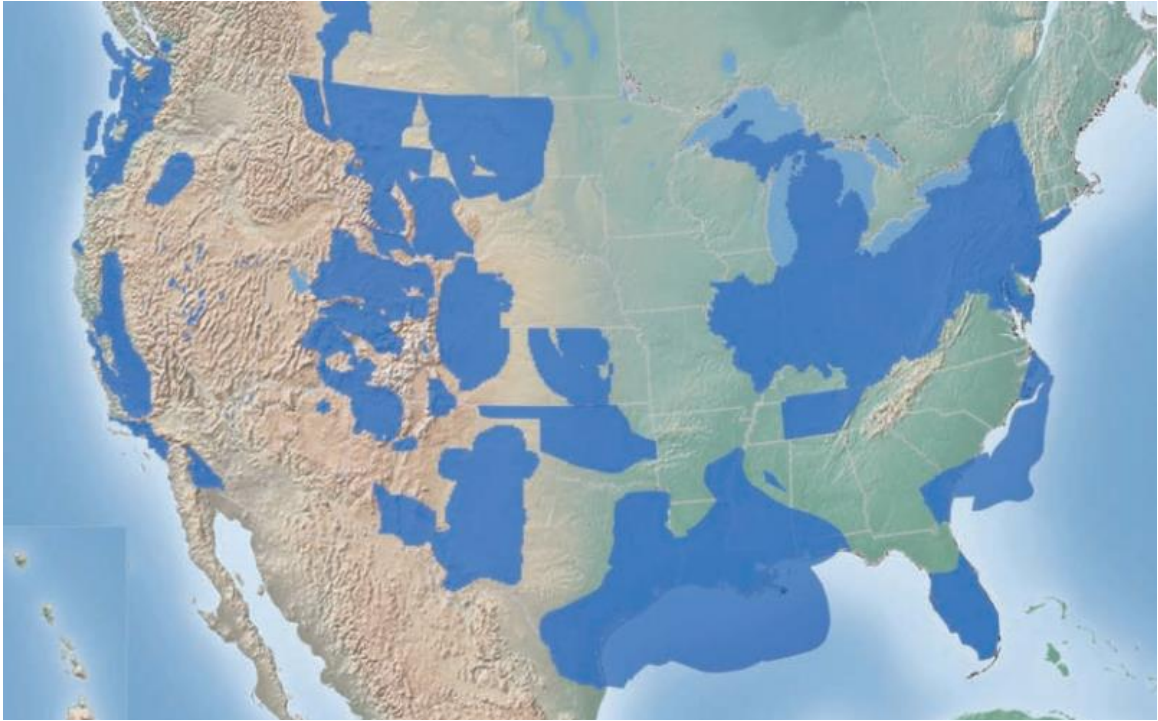


Figure 1.2 Map showing the location of assessed deep saline formations in North America (*DOE NETL, 2010*).

A deep saline aquifer consists of porous rock and is originally saturated with brine. When CO₂ is injected into the formation, a CO₂ plume forms and brine is displaced away from the injection well. Due to the large density difference between CO₂ and brine (ranging from 250 – 750 kg/m³), CO₂ will concentrate at the top of the aquifer and move away from the injection well along the caprock. One major concern of CCS is leakage of CO₂ from the intended storage formation through caprock imperfections such as fractures, faults and abandoned wells. Any significant leakage of CO₂ to the atmosphere negates the greenhouse gas emission reduction of CCS, and subsurface leakage also has the potential to acidify overlying aquifers. Another concern of CCS is induced seismicity. When CO₂ is injected, the injection-induced elevated fluid pressure in the formation could trigger seismic events and create leakage pathways (*Zoback and Gorelick, 2012*).

However, site characterization and proper site selection should reduce the risk of seismic events (Oldenburg, 2012; Birkholzer et al., 2012).

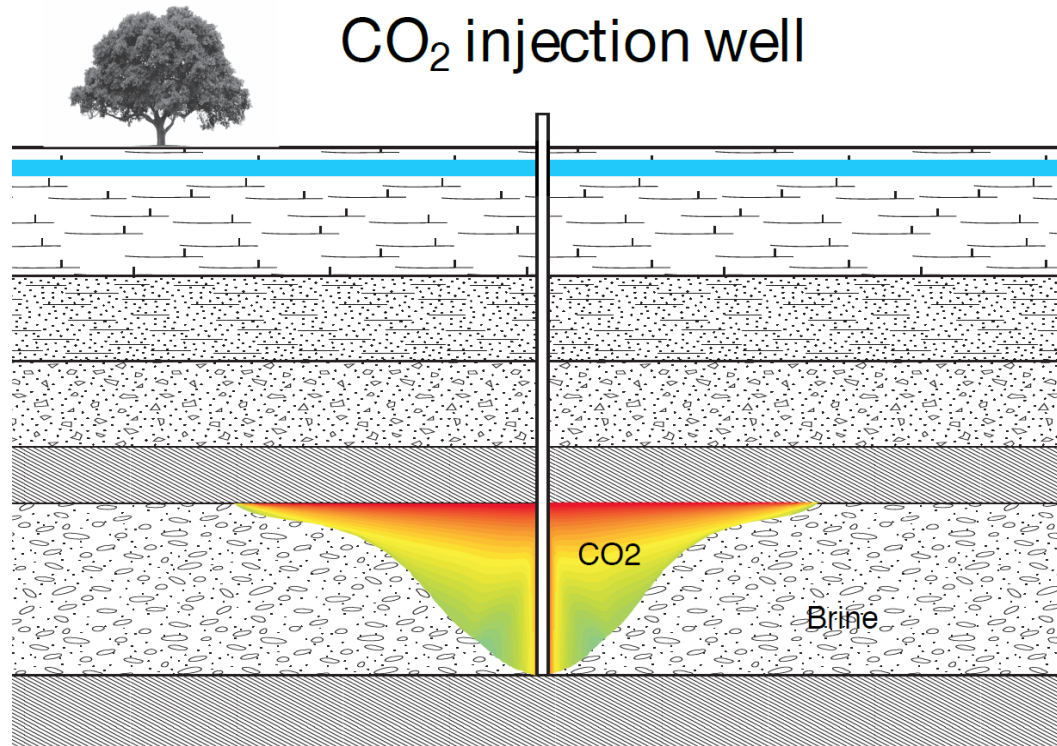


Figure 1.3 Schematic cross-section of a geological storage site showing that injected CO₂ forms a plume and displaces brine in a deep saline aquifer (modified from Bo Guo, 2016).

Accurate modeling of CO₂ migration in geological formations provides critical information about underground CO₂ storage capacity, spatial location of CO₂ plumes over time, and fluid pressure evolution in the formation. Estimates of underground CO₂ storage capacity allow policymakers to make informed decision regarding climate change mitigation strategies. The prediction of fluid migration, both CO₂ and brine, and pressure response in the subsurface allow lawmakers to create regulations to decrease the risk of CO₂ leakage, for instance through precautionary measures, such as plugging and intensive monitoring of abandoned hydrocarbon wells within a certain spatial range of the

CO₂ injection well. Modeling of CO₂ injection in geological formations is thus a vitally important part of the geological storage of CO₂ and the CCS process.

This thesis focuses on CO₂ injection into fractured geological formations, which are geological formations with natural and artificially induced fractures. According to the *U.S. Department of Energy* (2010), carbon dioxide enhanced oil recovery—the operation that injects CO₂ into declining hydrocarbon reservoirs to produce more of the remaining oil—has been pursued in the U.S. since 1970s and produced over 240,000 barrels of oil per day in the U.S. in 2008. Many of those oil and gas reservoirs are carbonate reservoirs and are highly fractured. Selected sites for CO₂ injection and storage can also have fractured formations. For example, the In Salah CO₂ storage project, an industrial scale demonstration project in Algeria, injected and stored CO₂ into fractured sandstones (*White et al.*, 2014).

Chapter 2

Background

The goal of modeling CO₂ migration in a geological formation is to obtain the spatial distribution of fluid pressure and fluid saturation in the domain throughout the simulation period. This goal can be achieved by using numerical models that solve the governing equations for multi-phase flow in porous media. Two types of modeling approaches, the three-dimensional (3D) model and the vertical-equilibrium (VE) model, can be used to solve for saturation and pressure variables in the domain of interest. The 3D model solves three-dimensional governing equations to obtain pressure and saturation in each grid cell in the domain. These spatially-distributed variables are called fine-scale pressure and saturation in this thesis.

The VE model assumes rapid and complete CO₂ and brine segregation in the vertical direction with a resulting vertical equilibrium (hydrostatic) fluid pressure distribution. In the VE approach, the three-dimensional governing equations are integrated in the vertical direction (i.e., perpendicular to the bedding plane of the formation) and become two-dimensional (horizontal) equations. The governing equations of the VE approach are written in terms of a reference pressure (e.g., at the bottom of the formation) and a vertically averaged saturation; these variables are termed coarse-scale pressure and

saturation in the context of VE models. Fine-scale pressure and saturation are reconstructed from the coarse-scale pressure and saturation using relatively simple expressions based on the assumed hydrostatic vertical pressure profiles. The VE modeling approach is much more computationally efficient than the 3D approach due to the dimension reduction from vertical integration. Both 3D and VE models have been applied extensively to model CO₂ injection and migration in geological formations in the past decade. For more details on modeling approaches applied to geologic carbon storage, the reader is referred to *Bandilla et al.* (2015).

In this chapter, I will first derive governing equations for the 3D model and the VE model. I will then discuss the methodology of modeling a fractured geological formation. Two types of models, namely dual-porosity and dual-permeability models, are discussed. Lastly, I will review the governing equations for dual-permeability Full-3D model and dual-porosity Full-3D model.

2.1 Three-Dimensional Governing Equations for Fluid Flow in Porous Media

A 3D model solves three-dimensional governing equations for pressure and saturation of the two fluid phases. Specifically, we are interested in obtaining four variables: CO₂ pressure p_c , brine pressure p_b , CO₂ saturation s_c , and brine saturation s_b . In order to solve for these four variables, four equations are needed. The following four equations are used in this thesis:

1. Pressure equation
2. Transport equation
3. Pressure relationship
4. Saturation relationship

Derivations of these four equations are shown below. The mass balance equations for the two fluid phases are given by

$$\frac{\partial (\rho_\alpha \phi s_\alpha)}{\partial t} + \nabla \cdot (\rho_\alpha \mathbf{u}_\alpha) = \rho_\alpha \psi_\alpha \quad (1)$$

where α identifies the fluid phase, with $\alpha = c, b$ to indicate CO₂ or brine, respectively, ρ_α is the density of CO₂ or brine, ϕ is the porosity of the geological formation, s_α is the saturation of CO₂ or brine, $\mathbf{u}_\alpha$ is the volumetric flux vector of CO₂ or brine in the formation, and ψ_α is the volumetric source or sink of CO₂ or brine in the formation. The time derivative can be written as follows

$$\frac{\partial (\rho_\alpha \phi s_\alpha)}{\partial t} = \rho_\alpha \phi \frac{\partial s_\alpha}{\partial t} + s_\alpha \rho_\alpha c_\alpha \frac{\partial p_\alpha}{\partial t} \quad (2)$$

where c_α is the total compressibility coefficient associated with phase α and p_α is the fluid pressure of phase α . c_α accounts for fluid and rock compressibility and $c_\alpha \equiv$

$\frac{\phi}{\rho} \frac{d\rho_\alpha}{dp_\alpha} + \frac{d\phi}{dp_\alpha}$. Combining equations (1) and (2) gives

$$\rho_\alpha \phi \frac{\partial s_\alpha}{\partial t} + s_\alpha \rho_\alpha c_\alpha \frac{\partial p_\alpha}{\partial t} + \nabla \cdot (\rho_\alpha \mathbf{u}_\alpha) = \rho_\alpha \psi_\alpha \quad (3)$$

When the spatial derivative of density is negligible, a “slight compressibility” assumption can be introduced to the system, allowing us to keep the time derivative of density while neglecting the spatial derivatives of density. Dividing through by the

density, ρ_α , and summing the two phase equations represented by equation (3), we obtain the pressure equation.

$$\phi \frac{\partial s_c}{\partial t} + \phi \frac{\partial s_b}{\partial t} + s_c c_c \frac{\partial p_c}{\partial t} + s_b c_b \frac{\partial p_b}{\partial t} + \nabla \cdot \mathbf{u}_c + \nabla \cdot \mathbf{u}_b = \psi_c + \psi_b \quad (4)$$

Because saturation of two phases sum to unity, the pressure equation can be written as

$$s_c c_c \frac{\partial p_c}{\partial t} + s_b c_b \frac{\partial p_b}{\partial t} + \nabla \cdot \mathbf{u}_c + \nabla \cdot \mathbf{u}_b = \psi_c + \psi_b \quad (5)$$

The CO₂ mass balance equation is chosen as the transport equation:

$$\phi \frac{\partial s_c}{\partial t} + s_c c_c \frac{\partial p_c}{\partial t} + \nabla \cdot \mathbf{u}_c = \psi_c \quad (6)$$

Volumetric flux of phase α can be obtained from the two-phase extension of Darcy's Law

$$\mathbf{u}_\alpha = -\frac{k_{r,\alpha} \mathbf{k}}{\mu_\alpha} (\nabla p_\alpha - \rho_\alpha \mathbf{g}) \quad (7)$$

where $k_{r,\alpha}$ is the relative permeability of CO₂ or brine, $\mathbf{k}$ is the intrinsic permeability tensor of the formation, μ_α is the viscosity of CO₂ or brine, and $\mathbf{g}$ is the gravity acceleration vector. Relative permeability is calculated using the Brooks-Corey model (Brooks and Corey, 1964)

$$k_{r,c} = (1 - s_e)^2 \left(1 - s_e^{\frac{2+\lambda}{\lambda}} \right) \quad (8)$$

and

$$k_{r,b} = s_e^{\frac{2+3\lambda}{\lambda}} \quad (9)$$

where

$$s_e = \frac{s_b - s_b^{res}}{1 - s_b^{res} - s_c^{res}} \quad (10)$$

and s_c^{res} and s_b^{res} are the residual saturations for CO₂ and brine, respectively, and λ is an empirical factor that represents the distribution of pore sizes. The pressure difference

between two fluid phases is defined as the capillary pressure, p^{cap} and it is assumed to be a function of fluid saturation. So there is a pressure relationship

$$p_c = p_b + p^{cap}(s_c) \quad (11)$$

Due to geometric constraints, the two fluid saturations needs to sum to unity, so that the saturation relationship states

$$s_b + s_c = 1 \quad (12)$$

In sum, we need to solve equations (5), (6), (11), and (12) with the substitution of equation (7) in order to obtain pressure of CO₂ and brine and saturation of CO₂ and brine.

2.2 Vertically-Integrated Governing Equations

Unlike a 3D model, a VE model assumes rapid and complete buoyant segregation of the two fluid phases. The resulting vertical equilibrium fluid pressure distribution allows relatively simple integration of the governing equations in the vertical direction. This reduction in dimensionality makes VE models computationally efficient but restricts their applicability to geological formations with relatively large vertical permeability.

Simulation results from the VE model shows excellent match with results from the 3D model when the permeability of the formation is on the order of 100 millidarcies (1 mD = $1 \times 10^{-15} \text{ m}^2$) or higher (*Court et al.*, 2012). Integration of equation (1) in the vertical direction from the bottom of the formation, ξ_B , to the top of the formation, ξ_T , gives

$$\int_{\xi_B}^{\xi_T} \left[\frac{\partial}{\partial t} (\rho_\alpha \phi s_\alpha) + \nabla \cdot (\rho_\alpha \mathbf{u}_\alpha) \right] dz = \int_{\xi_B}^{\xi_T} \rho_\alpha \psi^\alpha dz \quad (13)$$

Definitions for compressibility are given below

$$d\rho_\alpha = \frac{d\rho_\alpha}{dp} dp = \rho_\alpha c_\alpha dp \quad (14)$$

and

$$d\phi = \frac{d\phi}{dp} dp = c_\phi dp \quad (15)$$

where $c_\alpha \equiv \frac{1}{\rho_\alpha} \frac{d\rho_\alpha}{dp}$ and $c_\phi \equiv \frac{d\phi}{dp}$ are the compressibility coefficients of the fluid phases and the rock, respectively. Compressibility coefficients are assumed to be constants. With equation (14) and equation (15), the first term in the left hand side of equation (13) can be expanded as

$$\int_{\xi_B}^{\xi_T} \frac{\partial}{\partial t} (\rho_\alpha \phi S_\alpha) dz = \Phi S_\alpha \rho_\alpha c_\alpha \frac{\partial P_\alpha}{\partial t} + \rho_\alpha S_\alpha c_\phi \frac{\partial P_\alpha}{\partial t} + \rho_\alpha \Phi \frac{\partial S_\alpha}{\partial t} \quad (16)$$

where $\Phi \equiv \int_{\xi_B}^{\xi_T} \phi dz$ is the vertically-integrated porosity, P_α is the coarse-scale pressure of CO₂ or brine, and S_α is the coarse-scale saturation of CO₂ or brine. The terms “vertically-integrated” and “coarse-scale” are used interchangeably when describing a variable in this thesis. Note also that, in this thesis, fine-scale variables are denoted by lowercase characters and coarse-scale variables are denoted by uppercase characters. Using the “slight compressibility” assumption, the second left hand side term of equation (13) is

$$\int_{\xi_B}^{\xi_T} \nabla \cdot (\rho_\alpha \mathbf{u}_\alpha) dz = \rho_\alpha (\nabla_{\parallel} \cdot \mathbf{U}_\alpha + \psi_T^\alpha - \psi_B^\alpha) \quad (17)$$

where subscript “ $\parallel$ ” stands for vector components in the (x,y) plane, $\mathbf{U}_\alpha$ is the vertically-integrated volumetric flux vector of CO₂ or brine, and ψ_T^α and ψ_B^α are the source or sink terms representing CO₂ or brine flux entering or exiting the formation from the top and from the bottom, respectively.

Additional vertically-integrated (coarse-scale) variables are defined below. $\mathbf{K}$ is the vertically-integrated permeability tensor. $\mathbf{G}$ is the coarse-scale gravity term where $\mathbf{G} = \mathbf{e}_\parallel \cdot \mathbf{g} + (\mathbf{g} \cdot \mathbf{e}_z) \nabla_\parallel \xi_B$, $\mathbf{e}_\parallel = (e_x, e_y)^T$, and e is the unit vector. Λ_α is the coarse-scale mobility and is a function of fine-scale saturation (and λ_α is the fine-scale mobility, which is a function of fine-scale saturation).

$$\mathbf{U}_\alpha = -\mathbf{K} \Lambda_\alpha \cdot (\nabla_\parallel P_\alpha - \rho_\alpha \mathbf{G}) \quad (18)$$

$$\mathbf{K} \equiv \int_{\xi_B}^{\xi_T} \mathbf{k}_\parallel dz \quad (19)$$

$$\Lambda_\alpha \equiv \frac{1}{\mathbf{K}} \int_{\xi_B}^{\xi_T} \mathbf{k}_\parallel \lambda_\alpha(s_\alpha) dz \quad (20)$$

$$\lambda_\alpha(s_\alpha) \equiv \frac{k_{r,\alpha}}{\mu_\alpha} \quad (21)$$

For systems with large permeability, often the effects of fine-scale capillary pressure are small, and in some cases they can be ignored. This results in a capillary transition zone that shrinks to zero, corresponding to the so-called sharp interface assumption, which neglects fine-scale capillary pressure. The expressions for the coarse-scale mobility then simplify to the following

$$\Lambda_b = \left(1 - \frac{S_c}{1 - S_b^{res}}\right) \frac{1}{\mu_b} \quad (22)$$

$$\Lambda_c = \frac{S_c}{1 - S_b^{res}} \frac{k_{r,c}^*}{\mu_c} \quad (23)$$

where $k_{r,c}^*$ is the values of $k_{r,c}$ when $s_b = s_b^{res}$. Since $s_b^{res} = 0$ in this thesis, $k_{r,c}^* = 1$.

The derivations above give us expressions of the left hand side of equation (13) in terms of coarse-scale variables. Next let us look at the right hand side of equation (13).

The first term is:

$$\int_{\xi_B}^{\xi_T} \rho_\alpha \psi^\alpha dy = \rho_\alpha \Psi^\alpha \quad (24)$$

where Ψ^α is the vertically-integrated source or sink of CO₂ or brine in the domain. In sum, the vertically-integrated mass balance equation for fluid phases is shown below.

Essentially, equation (25) is the vertically-integrated version of equation (1).

$$\begin{aligned} \Phi \rho_\alpha \frac{\partial S_\alpha}{\partial t} + \Phi S_\alpha \rho_\alpha c_\alpha \frac{\partial P_\alpha}{\partial t} + S_\alpha \rho_\alpha c_\phi \frac{\partial P_\alpha}{\partial t} + \rho_\alpha \nabla_\parallel \cdot \mathbf{U}_\alpha \\ = \rho_\alpha \Psi^\alpha - \rho_\alpha \psi_T^\alpha + \rho_\alpha \psi_B^\alpha \end{aligned} \quad (25)$$

Dividing through by the density, ρ_α , and summing the two vertically-integrated phase equations represented by equation (25), we obtain the coarse-scale pressure equation for the domain.

$$(\Phi c_c + c_\phi) S_c \frac{\partial P_c}{\partial t} + (\Phi c_b + c_\phi) S_b \frac{\partial P_b}{\partial t} + \nabla_\parallel \cdot (\mathbf{U}_c + \mathbf{U}_b) = \Psi^c + \Psi^b \quad (26)$$

The vertically-integrated mass balance equation of CO₂ is chosen as the coarse-scale transport equation

$$\Phi \frac{\partial S_c}{\partial t} + (\Phi c_c + c_\phi) S_c \frac{\partial P_c}{\partial t} + \nabla_\parallel \cdot \mathbf{U}_c = \Psi^c \quad (27)$$

The difference between the two coarse-scale phase pressures is called “pseudo-capillary pressure” and is defined at the depth of the coarse-scale phase pressure. The pressure relationship is shown below

$$P_c = P_b + P^{cap}(S_c) \quad (28)$$

The pseudo-capillary pressure is a function of the coarse-scale saturation. From the sharp interface assumption, P^{cap} can be expressed as: $P^{cap} = P_{entry}^{cap} + (1 - S_c)H(\rho_c - \rho_b)g$ where H is the height of the formation. Lastly, the coarse-scale saturation relationship gives

$$S_b + S_c = 1 \quad (29)$$

In sum, to obtain coarse-scale pressure and saturation of CO₂ and brine in the domain, we need to solve equations (26) – (29) with the substitution of equation (18). Notice that equations (26) – (27) are two-dimensional equations and solving them takes much less computational effort as compare to solving their three-dimensional counterparts in the 3D model, which are equation (5) and equation (6).

2.3 Models for a Dual-Continuum System

A fractured geological formation is a geological formation with natural and/or artificially induced fractures, which are extensive, abundant, and highly connected with each other. As a result, the fractures themselves can be treated as a geological formation with large permeability and low porosity and can be modeled as a continuum. The portion of the formation that is not fractures (i.e., the rock matrix) is modeled as a second continuum, defined by small permeability and large porosity. In other words, a fractured geological formation can be modeled as an overlap of two continua, one representing the fractures and the other representing the rock matrix. Modeling approaches that treat both fractures and matrix as continua are therefore called dual-continuum approaches.

In dual-continuum approaches the two most common ways to conceptualize the rock matrix are: the dual-porosity approach and the dual-permeability approach. The dual-porosity model was developed by *Warren and Root (1963)* and this approach models the rock matrix portions of the fractured formation as identical rectangular parallelepipeds with homogeneous and isotropic rock properties. Those matrix blocks are not directly connected to each other and they only exchange fluids (brine and CO₂) with fractures. As a result, there is no large-scale flow within the rock matrix domain and, therefore, the rock matrix effectively serves as a source or sink of fluids that flow in the fractures. There have been extensions of the dual-porosity model such as *Kazemi et al. (1976)* and *Thomas et al. (1983)*, but in this thesis we only discuss the original *Warren and Root (1963)* dual-porosity approach.

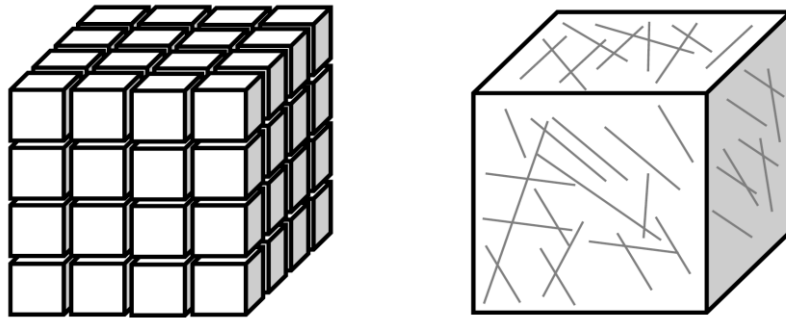


Figure 2.1 Schematic graphs showing the dual-porosity conceptualization (left) and the dual-permeability conceptualization (right) (taken from *Bo Guo, 2016*).

The dual-permeability model was developed by *Barenblatt et al. (1960)* and this approach models the rock matrix portions of the fractured formation as a continuum with permeability and porosity different from fractures. Flow can thus occur within the rock matrix domain. The transfer of fluid mass between rock matrix and fractures is a key process in both dual-porosity and dual-permeability models and is represented by a mass

transfer function. We adopted the mass transfer function for the dual-porosity model from *Ramirez et al. (2009)* and adopted the one for the dual-permeability model from *Sonier et al. (1988)*. Recent reviews of various versions of the mass transfer functions can be found in *Ramirez et al. (2009)* and *Al-Kobaisi et al. (2009)*.

The primary focus of this thesis is on dual-permeability models. “Full-3D model” and “VE-3D model” in the text mean dual-permeability models, unless specified otherwise. Developing dual-porosity models shares many of the common steps as developing dual-permeability models. In this thesis, discussions of dual-porosity models are limited to Section 2.4.2 and Section 3.1.2, which show dual-porosity governing equations, and Section 4.1, which shows dual-porosity simulation results.

In the following sections, two types of dual-continuum models, the Full-3D and VE-3D models, are discussed. The Full-3D model has been used to model CO₂ injection in fractured reservoirs for decades. The VE-3D model is an innovative model recently developed by the author and the research group, and is the main focus of this thesis.

2.4 Governing Equations for a Full-3D Model

2.4.1 Dual-Permeability Model

A Full-3D model uses the 3D model in both fracture and rock matrix domains. Fluid exchanges between the fracture and rock matrix domains are represented by mass transfer terms that are included in the governing equations of the two domains. Mass transfer terms are expressed as functions of variables from both fracture and rock matrix domains

and account for physical mechanisms such as viscous forces, capillary pressure, and gravity drainage.

The mass transfer terms are added to the governing equations for the fracture and rock matrix domains and take the form of source/sink fluxes. Because the mass transfer terms, q_c^{f-m} and q_b^{f-m} , are defined as mass fluxes from fractures to rock matrix, signs in front of them are minus in the fracture domain and are plus in the rock matrix domain. The pressure and transport equations for the two domains of the Full-3D model are shown below. Equations (30) and (31) are for the fractures and equations (32) and (33) are for the rock matrix. The pressure and saturation relationships are the same as equations (11) and (12). Variables with superscript f are from the fracture domain and ones with superscript m are from the rock matrix domain.

$$s_c^f c_c^f \frac{\partial p_c^f}{\partial t} + s_b^f c_b^f \frac{\partial p_b^f}{\partial t} + \nabla \cdot \mathbf{u}_c^f + \nabla \cdot \mathbf{u}_b^f = \psi_c^f + \psi_b^f - q_c^{f-m} - q_b^{f-m} \quad (30)$$

$$\phi^f \frac{\partial s_c^f}{\partial t} + s_c^f c_c^f \frac{\partial p_c^f}{\partial t} + \nabla \cdot \mathbf{u}_c^f = \psi_c^f - q_c^{f-m} \quad (31)$$

$$s_c^m c_c^m \frac{\partial p_c^m}{\partial t} + s_b^m c_b^m \frac{\partial p_b^m}{\partial t} + \nabla \cdot \mathbf{u}_c^m + \nabla \cdot \mathbf{u}_b^m = \psi_c^m + \psi_b^m + q_c^{f-m} + q_b^{f-m} \quad (32)$$

$$\phi^m \frac{\partial s_c^m}{\partial t} + s_c^m c_c^m \frac{\partial p_c^m}{\partial t} + \nabla \cdot \mathbf{u}_c^m = \psi_c^m + q_c^{f-m} \quad (33)$$

The mass transfer functions that are suitable for the dual-permeability model are adopted from *Sonier et al.* (1988)

$$q_c^{f-m} = \sigma k^m \left[\frac{k_{r,c}^f}{\mu_c} \delta_c^f + \frac{k_{r,c}^m}{\mu_c} (1 - \delta_c^f) \right] \left[p_b^f - p_b^m + (p^{cap})^f - (p^{cap})^m + \frac{\sigma_z}{\sigma} \Delta \rho g \left(\frac{s_c^f - s_{cr}^f}{1 - s_{cr}^f} - \frac{s_c^m - s_{cr}^m}{1 - s_{cr}^m} \right) l_z \right] \quad (34)$$

$$q_b^{f-m} = \sigma k^m \left[\frac{k_{r,b}^f}{\mu_b} \delta_b^f + \frac{k_{r,b}^m}{\mu_b} (1 - \delta_b^f) \right] \left[p_b^f - p_b^m + \frac{\sigma_z}{\sigma} \Delta \rho g \left(\frac{s_b^f - s_{br}^f}{1 - s_{br}^f} - \frac{s_b^m - s_{br}^f}{1 - s_{br}^f} \right) l_z \right] \quad (35)$$

where σ is the shape-factor for the rock matrix blocks and $\sigma = 4 \left(\frac{1}{l_x^2} + \frac{1}{l_y^2} + \frac{1}{l_z^2} \right)$, σ_z is the component of σ considering only the vertical direction and $\sigma_z = \frac{4}{l_z^2}$, δ_α^f is used for upstream weighting of the relative permeability and δ_α^f equals to 1 when the flux of CO₂ or brine is going from the fracture domain to the rock matrix domain and equals to 0 otherwise, l_x is the length of a matrix block, l_z is the height of a matrix block, and $s_{\alpha r}^f$ and $s_{\alpha r}^m$ are residual saturation for CO₂ or brine in the fracture domain and in the rock matrix domain, respectively.

Mass transfer functions for dual-permeability models—equations (34) and (35)—consider three physical driving forces: viscous force represented by the difference of the pressure of the same phase in the two domains, capillary force represented by the difference of the capillary pressure in the two domains, and gravitational force, represented by the difference in local fluid saturation in the two domains.

2.4.2 Dual-Porosity Model

A dual-porosity model can be viewed as a dual-permeability model with no permeability in the rock matrix domain. There are two differences between a Full-3D dual-porosity model and a Full-3D dual-permeability model. Firstly, there is no large-scale flow in the rock matrix domain, and therefore no equations for phase pressures are needed for the

rock matrix domain. Secondly, the dual-porosity model uses mass transfer functions different from dual-permeability model. The equation that shows the change of mass in rock matrix over time is

$$\phi^m \frac{\partial s_c^m}{\partial t} = \psi_c^m + q_c^{f-m} \quad (36)$$

Mass transfer functions for dual-porosity models take into account capillary pressure and gravity drainage. The mass transfer functions are adopted from *Ramirez et al.*, (2009). In the implementation, we assume the fluids are incompressible. Because rock matrix only works as a storage space for fractures, the transfer fluxes of CO₂ and brine must sum to zero to make the volume balance.

$$q_c^{f-m} = \sigma k^m \frac{\lambda_c^f \lambda_b^m}{\lambda_c^f + \lambda_b^m} \left\{ -[(p^{cap})^m - (p^{cap})^f] + \frac{\sigma_z}{\sigma} \Delta \rho g \left(\frac{s_c^f - s_{cr}^f}{1 - s_{cr}^f} - \frac{s_c^m - s_{cr}^f}{1 - s_{cr}^f} \right) l_z \right\} \quad (37)$$

$$q_b^{f-m} = -q_c^{f-m} \quad (38)$$

where λ_α^f is the mobility of phase α in fractures and $\lambda_\alpha^f = \frac{k_{r,\alpha}^f}{\mu_\alpha}$.

Chapter 3

Methods

In this chapter, I will first show governing equations and mass transfer functions for the VE-3D dual-permeability model and the VE-3D dual-porosity model. I will then describe the domain discretization setup and the numerical methods that are used to discretize the governing equations and I will discuss the boundary conditions and initial conditions of the model. After that, I will describe the IMPES method, which illustrates the steps of solving discrete governing equations for pressure and saturation in the domains. Lastly, I will summarize the complexities and challenges in implementing the Full-3D code and the VE-3D code.

3.1 Governing Equations for a VE-3D Model

3.1.1 Dual-Permeability Model

A VE-3D model for fractured geological formations uses a VE model in the fracture domain and a 3D model in the rock matrix domain. A VE model assumes rapid and complete buoyant segregation of the two fluid phases and resulting vertical equilibrium pressure distribution. This assumption is likely appropriate for the fracture domain because the permeability of the fractures is expected to be high and CO_2 in the fractures

thus segregates relatively quickly in the vertical direction due to the large density difference between CO₂ and brine. Brine will stay beneath the CO₂ plume.

Because of the large pores in the fractures, capillary transition zones are expected to be small and the use of sharp interface assumption in the VE model is appropriate. We add vertically-integrated mass transfer terms to the vertically-integrated governing equations of the VE model. The coarse-scale pressure equation for the fracture domain becomes:

$$\begin{aligned} & \left(\Phi^f c_c^f + c_\phi^f \right) S_c^f \frac{\partial P_c^f}{\partial t} + \left(\Phi^f c_b^f + c_\phi^f \right) S_b^f \frac{\partial P_b^f}{\partial t} + \nabla_{\parallel} \cdot (\mathbf{U}_c^f + \mathbf{U}_b^f) \\ & = (\Psi^c)^f + (\Psi^b)^f - Q_c^{f-m} - Q_b^{f-m} \end{aligned} \quad (39)$$

where $Q_\alpha^{f-m} = \int_{\xi_B}^{\xi_T} q_\alpha^{f-m} dz$. The transport equation for the fracture domain is:

$$\Phi^f \frac{\partial S_c^f}{\partial t} + \left(\Phi^f c_c^f + c_\phi^f \right) S_c^f \frac{\partial P_c^f}{\partial t} + \nabla_{\parallel} \cdot \mathbf{U}_c^f = (\Psi^c)^f - Q_c^{f-m} \quad (40)$$

The rock matrix domain of the VE-3D model is treated in the same way as in the Full-3D model. We solve for fine-scale pressure and saturation of CO₂ and brine in each grid cell in the domain. As a result, the pressure equation and transport equation for the rock matrix domain are equations (32) and (33).

Mass transfer functions of CO₂ and brine are essentially the same in the VE-3D model and in the Full-3D model. We, however, need to integrate the mass transfer functions in the vertical direction in order to include them in the fractures' governing equations. Expressions for vertically-integrated mass transfer functions are shown below.

$$\begin{aligned}
Q_c^{f-m} &= \int_{\xi_B}^{\xi_T} q_c^{f-m} dz \\
&= \int_{\xi_B}^{\xi_T} \sigma k^m \left[\frac{k_{r,c}^f}{\mu_c} \delta_c^f + \frac{k_{r,c}^m}{\mu_c} (1 - \delta_c^f) \right] \left[p_b^f - p_b^m + (p^{cap})^f \right. \\
&\quad \left. - (p^{cap})^m + \frac{\sigma_z}{\sigma} \Delta \rho g \left(\frac{s_c^f - s_{cr}^f}{1 - s_{cr}^f} - \frac{s_c^m - s_{cr}^f}{1 - s_{cr}^f} \right) l_z \right] dz
\end{aligned} \tag{41}$$

$$\begin{aligned}
Q_b^{f-m} &= \int_{\xi_B}^{\xi_T} q_b^{f-m} dz \\
&= \int_{\xi_B}^{\xi_T} \sigma k^m \left[\frac{k_{r,b}^f}{\mu_b} \delta_b^f + \frac{k_{r,b}^m}{\mu_b} (1 - \delta_b^f) \right] \left[p_b^f - p_b^m \right. \\
&\quad \left. + \frac{\sigma_z}{\sigma} \Delta \rho g \left(\frac{s_b^f - s_{br}^f}{1 - s_{br}^f} - \frac{s_b^m - s_{br}^f}{1 - s_{br}^f} \right) l_z \right] dz
\end{aligned} \tag{42}$$

Since the coarse-scale pressure equation in the fracture domain—equation (39)—and pressure equation in the rock matrix domain—equation (32)—both contain fine-scale brine pressure in the fracture domain, p_b^f , the reconstruction of p_b^f from P_b^f becomes part of the pressure equations. Suppose the reference level for coarse-scale pressure is immediately above the bottom of the formation, and the positive z-direction is perpendicular to the bedding plane and pointing to the top of the formation, then the equation for CO₂ pressure in the plume is $p_c^f = P_c^f - \rho_c g z$ and the equation for brine pressure is $p_b^f = P_b^f - \rho_b g z$. Q_c^{f-m} in the coarse-scale transport equation is calculated by vertically summing q_c^{f-m} in the column that corresponds to Q_c^{f-m} 's location in the domain.

3.1.2 Dual-Porosity Model

In the fracture domain, a VE-3D dual-porosity model has the same pressure equation and transport equation as a VE-3D dual-permeability model does, which are equation (39) and equation (40). In the rock matrix domain, the governing equation of VE-3D dual-porosity model is the same as the one of Full-3D dual-porosity model, which is equation (36).

The mass transfer functions for Full-3D dual-porosity model and VE-3D dual-porosity model are the same but we need to integrate them in the vertical dimension in order to put them in the governing equations of the fractures. Expressions for vertically-integrated mass transfer functions are shown below. Coarse-scale mass transfer terms, Q_α^{f-m} , are handled in the same way as described in Section 3.1.1.

$$\begin{aligned}
 Q_c^{f-m} &= \int_{\xi_B}^{\xi_T} q_c^{f-m} dz \\
 &= \int_{\xi_B}^{\xi_T} \sigma k^m \frac{\lambda_c^f \lambda_b^m}{\lambda_c^f + \lambda_b^m} \left\{ -[(p^{cap})^m - (p^{cap})^f] \right. \\
 &\quad \left. + \frac{\sigma_z}{\sigma} \Delta \rho g \left(\frac{s_c^f - s_{cr}^f}{1 - s_{cr}^f} - \frac{s_c^m - s_{cr}^f}{1 - s_{cr}^f} \right) l_z \right\} dz
 \end{aligned} \tag{43}$$

$$Q_b^{f-m} = -Q_c^{f-m} \tag{44}$$

3.2 Implementation

In this section, I will discuss the code implementation of a Full-3D model and a VE-3D model. I will first introduce the setup of grid cells in the domain and then discuss the boundary conditions and initial conditions that are used in the model. After that I will discuss the IMPES method and summarize code implementation complexities.

Derivations of discrete governing equations for Full-3D model and VE-3D model are shown in the Appendix.

3.2.1 Domain Discretization

Fracture and rock matrix domains in this thesis are based on a typical geological formation and are slices in the (x,z) plane. This could be seen as modeling a slice perpendicular to a horizontal injection well. The y -dimension is neglected because there is no flow in the y -direction. The right graph of Figure 3.1 shows the discretization of the fracture and rock matrix domains and the mass transfer fluxes (blue arrows) that connect the two domains. The size of grid cells can vary across the domain but they are set to be the same in the test case. The domain discretization is the same for the VE and 3D models.

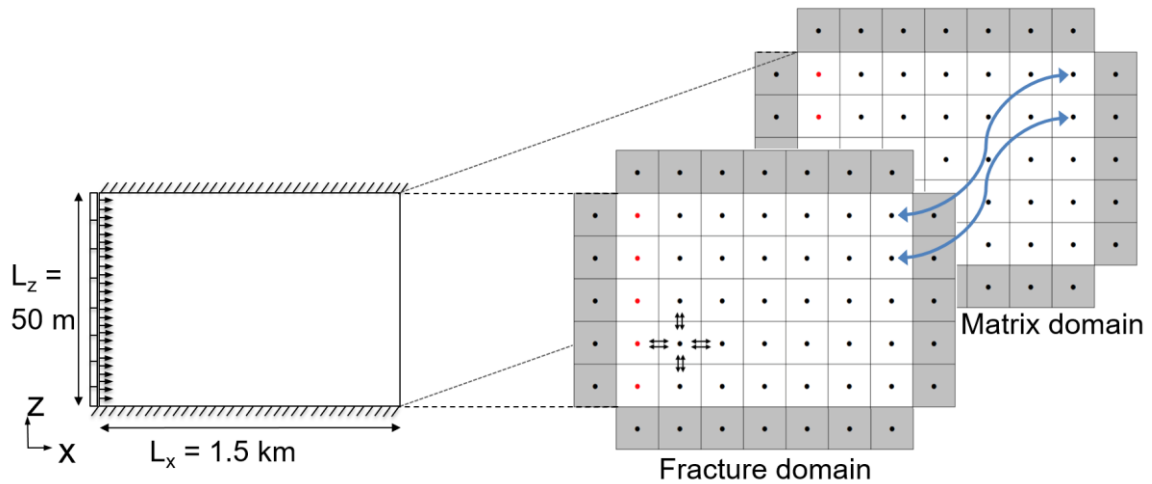


Figure 3.1 Picture showing the discretization of fracture and rock matrix domains into numerical domains.

There are two types of grid cells: cells in white are in the numerical domain while cells in grey are outside of the numerical domain. There are two types of white cells. The white cells that are adjacent to grey cells are called “boundary cells” because one or two

of their four edges constitutes the boundary of the numerical domain. White cells that are not “boundary cells” are called “in-domain cells”. Expressions of the governing equations are the same for all “in-domain cells” while they are different in “boundary cells” depending on boundary conditions. Grey cells that surround the numerical domain are called “ghost cells” and they are used to facilitate the discretization of the governing equations along the boundaries.

Each white cell is connected to its four adjacent cells as well as its counterpart in the other numerical domain by fluid fluxes. The permeability of the geological formation is defined in each grid cell and can vary in the numerical domain to model layers of heterogeneous rocks. However, the geological formation is assumed to be homogeneous in the test cases so the permeability is the same in all grid cells.

3.2.2 Boundary Conditions and Initial Conditions

The boundary of the numerical domain is defined at the outer boundaries of the “boundary cells”, instead of the centers of the “boundary cells”. In test case used in this study, there are no fluid fluxes going through the top and bottom boundaries of the numerical domain. This is realistic because the selected geological formation for CO₂ storage is likely to have impermeable caprocks on the top of it and beneath it. There is no flux going through the left boundary of the numerical domain due to the symmetry (i.e., the left and right sides of the injection well are exactly the same). The right boundary conditions are hydrostatic brine pressure and $\frac{dp^{cap}}{dx} = 0$. The latter condition means the saturation in a right “ghost cell” is the same as the saturation in the adjacent right

“boundary cell”. The right boundary conditions would hold in the test case of this thesis even though the fine-scale capillary pressure is set to zero in the test case.

The horizontal injection well would go through only one grid cell in the domain. However, in this thesis we inject constant CO₂ flux into all left “boundary cells” because this injection pattern makes the 3D model more consistent with the VE model, which has no representation in the z-direction. Before CO₂ injection starts, the geological formation contains no CO₂ and all pore spaces are occupied by brine. As a result, the initial conditions are saturation of brine equals to one and hydrostatic brine pressure in the domain.

3.2.3 IMPES Method

Because we are interested in how the CO₂ plume migrates and how the fluid pressure builds up in the geological formation with time, we discretize the temporal dimension and solve the governing equations for pressure and saturation variables in each time step throughout the simulation period. The Implicit Pressure and Explicit Saturation (IMPES) method is a common solution procedure and is used in both the Full-3D and VE-3D models.

For a Full-3D model, the IMPES method proceeds by first solving the pressure equations of both domains for brine pressure in each grid cell at the new time level. The nonlinear coefficients in the pressure equations are based on saturation values from the previous time step. As a result, brine pressure is evaluated implicitly and the coefficients are lagged in time to linearize the equations. CO₂ pressure in each grid cell of both domains at the new time level is then calculated from brine pressure using the pressure

relationship. Next, CO₂ pressure is used in the transport equations of both domains to calculate updated fluxes across all grid cell boundaries. Everything in the transport equations is known except saturation in the next time level. CO₂ saturation is thus solved explicitly and brine saturation is then calculated from the saturation relationship. With newly updated CO₂ and brine saturations, the nonlinear coefficients in the pressure equations and the capillary pressure terms are updated and the algorithm proceeds to the next time step.

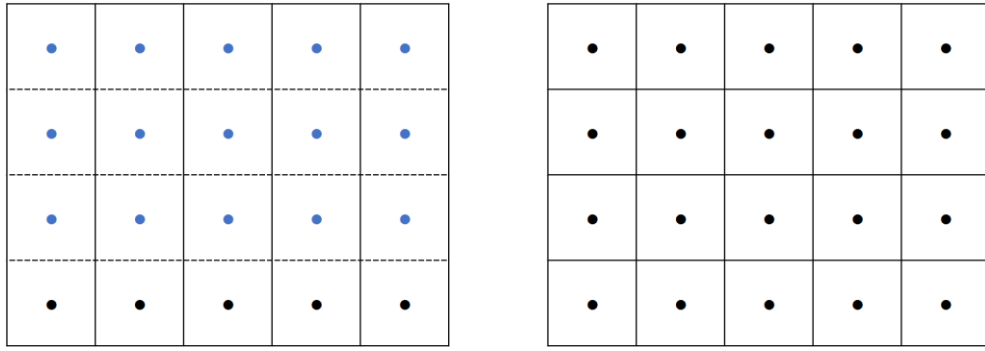


Figure 3.2 Pictures showing the discretized fracture domain (left) and the discretized rock matrix domain (right) in the x-z dimension in a VE-3D model. Pressure and saturation values at black and blue dots represent the values for the grid cell. Values at black dots are solved from governing equations. Values at blue dots are reconstructed from black dots in the fracture domain.

The IMPES algorithm for the VE-3D model, however, is more complicated and involves the reconstruction of fine-scale pressure and saturation from coarse-scale ones in each time step. In this thesis, the reference level for coarse-scale pressure is chosen to be the center of the bottom “boundary cells”. At the beginning of each time step, p_b^m in each grid cell of the rock matrix domain and P_b^f in each grid cell of the fracture domain (all black dots in Figure 3.2) are solved simultaneously from the pressure equation of the rock matrix domain and the vertically-integrated pressure equation of the fracture domain. p_c^m

in the rock matrix domain and P_c^f in the fracture domain are then calculated from p_b^m in the rock matrix domain and P_b^f in the fracture domain using the fine-scale and coarse-scale capillary pressure relationships, respectively. p_c^f in other grid cells of the fracture domain (blue dots in Figure 3.2) are then reconstructed from P_c^f in the fracture domain.

p_c^f and p_c^m in both domains and P_c^f in the fracture domain are used in the transport equation of the rock matrix domain and the vertically-integrated transport equation of the fracture domain. Then s_c^m in each grid cell of the rock matrix domain and S_c^f in each grid cell of the fracture domain are calculated. s_c^f values in all grid cells of the fracture domain are then reconstructed from S_c^f . s_b^f and s_b^m in both domains can then be calculated since s_c^f and s_c^m are known. Finally, we obtain values of p_c^f , p_b^f , s_c^f , s_b^f , p_c^m , p_b^m , s_c^m , and s_b^m in both domains in the current time step and the algorithm proceeds to the next time step.

3.2.4 Implementation Complexities

In the following, five complexities involved in implementing the Full-3D code and the VE-3D code are discussed. First, in both the Full-3D model and the VE-3D model, values of relative permeability of CO₂ or brine at the boundary of a grid cell are needed to calculate the fluid flux across that boundary, as indicated by equation (7). However, we only have values of relative permeability at the center of a grid cell. The method of upstream weighting is used to calculate relative permeability of CO₂ and brine at the grid cell boundary. At the beginning of each time step, the direction of CO₂ and brine fluxes across all adjacent grid-cells is determined by comparing the hydraulic heads in adjacent

cells. The relative permeability at the shared boundary of two adjacent grid cells is equal to the relative permeability at the center of the upstream grid cell. Relative permeability of CO₂ and brine in the mass transfer functions are also calculated using upstream weighting by making values of δ_{α}^f dependent on directions of the mass transfer fluxes.

Second, in the VE-3D model, the pressure equation of the rock matrix domain and the vertically-integrated pressure equation of the fracture domain need to be solved simultaneously because solving them sequentially leads to stability issues. Some coefficients in the computational matrix and some right hand side terms correspond to the vertically-integrated pressure equation and they only include coarse-scale variables. Other coefficients and right hand side terms are parts of the fine-scale pressure equation and they only include fine-scale variables.

Third, numbering of the grid cells in the domains would affect how fast the models run. In both the Full-3D and the VE-3D models, the grid cells are numbered from left to right and from bottom to top. In the Full-3D model, this numbering brings the convenience that non-zero terms are only located in the diagonals of the computational matrix, and the number of non-zero diagonals does not depend on domain discretization. No matter how fine the discretization is, there are seven non-zero diagonals in the computational matrix. As a result, I first assemble all non-zero diagonals and then place them into the computational matrix using the “sparse matrix” command in MATLAB®. In this way, the size of the computational matrix is greatly reduced and the running time becomes significantly shorter. In VE-3D model, this numbering means the number of non-zero diagonals of the computational matrix would increase with the number of grid cells in the domain. As a result, the “sparse matrix” command is not used in VE-3D

model and the computational matrix is assembled by directly placing items into the right location in each time step.

Fourth, the sizes of the computational matrices are different in the Full-3D model and the VE-3D model. Suppose there are 300 grid cells in horizontal direction and 100 grid cells in vertical direction. The size of the computational matrix is 60000×60000 ($60000 = 300 \times 100 + 300 \times 100$) in the Full-3D model and is 30300×30300 ($30300 = 300 \times 100 + 300$) in the VE-3D model. The smaller computational matrix size means that the VE-3D model takes significantly less computational effort than the Full-3D model.

Last but not least, the pressure reconstruction process of the VE-3D model needs special attention. The difference of phase pressure in the mass transfer functions is expressed in different forms depending on whether the grid cell that we are solving is in the CO₂ plume or is in brine. As a result, we need to first decide if a cell has non-zero CO₂ saturation before we enter its coefficients to the computational matrix. Moreover, the pressure and saturation reconstruction in the grid cell that has the CO₂-brine interface in it can be tricky. In my code, if the fine-scale saturation in the interface cell is larger than 0.5, then the cell is in the CO₂ plume. Otherwise the grid cell is in brine.

Chapter 4

Results and Discussion

As shown in Figure 4.1, test cases in this thesis use domains that are based on a typical geological formation and are slices in the (x,z) plane. The horizontal length of the domain is 1500 meters and is divided into 300 grid cells. The height of the domain is 50 meters and is divided into 100 grid cells. In total, 30000 grid cells discretize the domain. CO₂ is injected uniformly throughout the height of the domain at the left boundary.

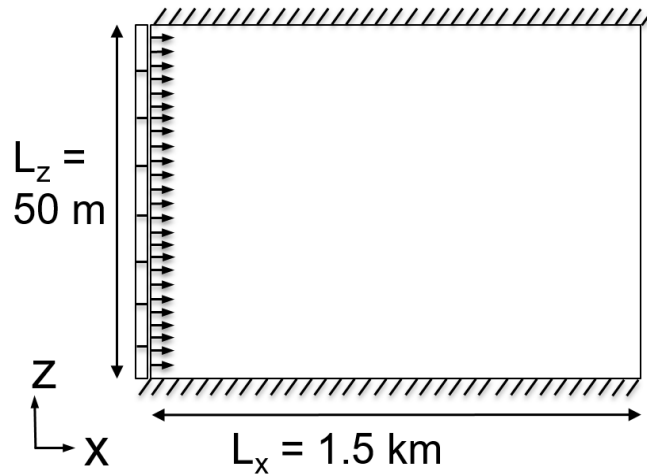


Figure 4.1 Picture showing the size of the geological formation that is used as the basis for test cases in this thesis.

The domains are assumed to be homogeneous. Unless specified otherwise, the permeability and porosity of the fracture domain is 1 Darcy ($1 \text{ D} = 10^3 \text{ mD}$) and 0.03, respectively, and the permeability and porosity of the rock matrix domain is 1 mD and

0.15, respectively. Capillary pressure and compressibility are neglected in all test cases for simplicity. In the Full-3D model, a Brooks-Corey function with $\lambda = 2$ and zero fluid residual saturation is used to calculate the relative permeability in both of its domains. In VE-3D model, a Brooks-Corey function is used in the rock matrix domain. In the fracture domain, the sharp interface model makes the types of the relative permeability curves unimportant because saturation is either zero or 1 after the integration in all grid cells other than the ones that contain the CO₂-brine interface. In this thesis, a linear relative permeability function is used in the fracture domain to fit the way interface cells are handled (described in Section 3.2.4). The injection rate is 1.8×10^{-3} kg CO₂ per second per meter in the y-direction. Density of CO₂ and brine is 710 kg/m³ and 1000 kg/m³, respectively. Viscosity of CO₂ and brine is 4.25×10^{-5} Pa·s and 3×10^{-4} Pa·s, respectively.

In Section 4.1, we compare Full-3D dual-porosity model with Full-3D dual-permeability model. After Section 4.1, we only look at dual-permeability models. Section 4.2.1 shows the comparison between VE-3D model and Full-3D model after 1 year and 5 years of continuous CO₂ injection. In Section 4.2.2, we compare results from the VE-3D model and the Full-3D model with fracture permeability ranging from 0.01 D to 100 D. Section 4.3 shows the comparison among VE-3D models when different forms of mass transfer functions are used. This can indicate which physical mechanism is playing a more important role in the fluid mass exchange between fractures and rock matrix.

4.1 Full-3D: Comparison between Dual-Porosity Model and Dual-Permeability Model

After injecting for 5 years, CO₂ saturation in the fracture and rock matrix domains of the Full-3D dual-porosity model and the Full-3D dual-permeability model are shown below. The dual-porosity model uses the same domain discretization, boundary conditions and initial conditions that are discussed in Section 3.2.

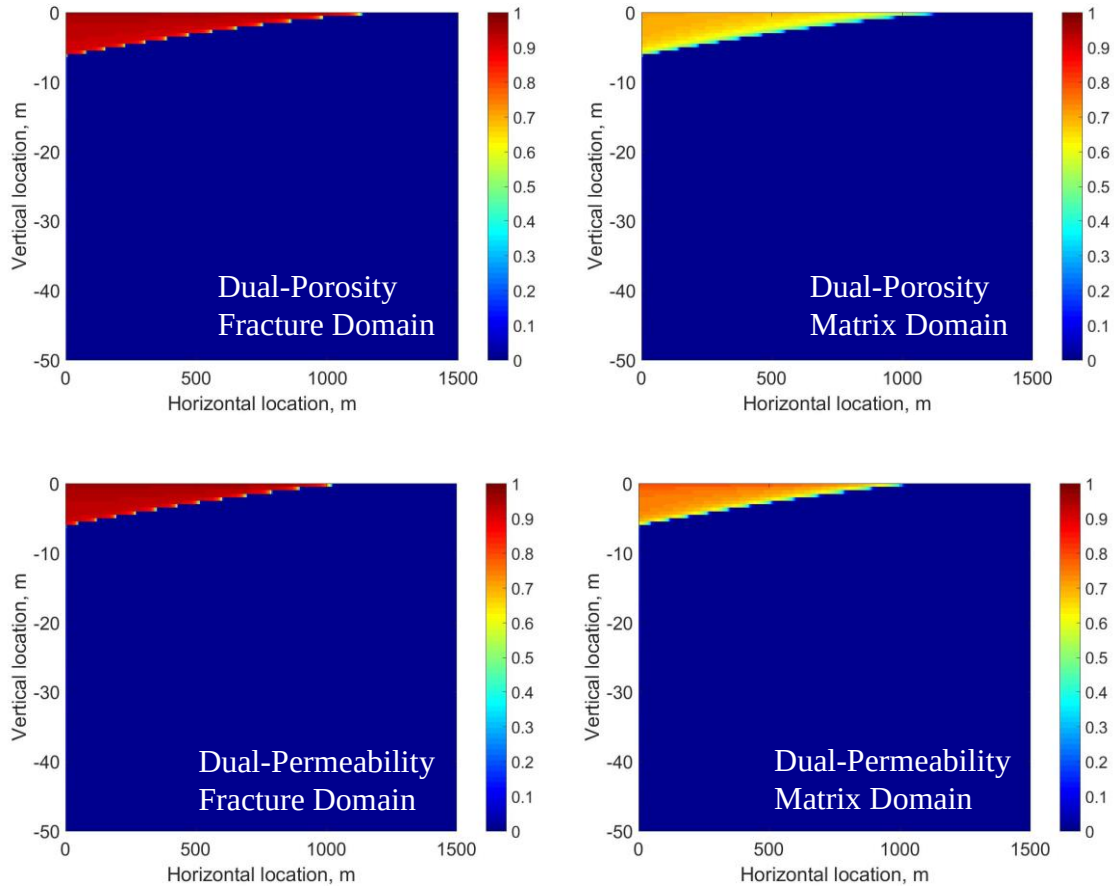


Figure 4.2 CO₂ saturation graphs from dual-porosity model (top) and dual-permeability model (bottom). Red color means high CO₂ saturation (low brine saturation) and blue color represents low CO₂ saturation (high brine saturation).

For the dual-porosity model, CO₂ is distributed evenly in the plume of the rock matrix domain because the model assumes no flow from one matrix block to another. For the dual-permeability model, the saturation variability in the CO₂ plume of the rock matrix domain is larger because CO₂ is slowly moving upward in the rock matrix and concentrates at the top. In the fracture domain, the dual-permeability model shows slightly larger CO₂ saturation and saturation variability in the plume due to the influence

from the rock matrix domain. For both domains, the plume sizes are smaller in the dual-permeability model because CO₂ is more concentrated in the plumes.

	Percentage of Injected CO ₂ Mass in the Fracture Domain	Percentage of Injected CO ₂ Mass in the Rock Matrix Domain
Dual-Porosity Model	23.09%	76.91%
Dual-Permeability Model	21.09%	78.91%

Table 4.1 Comparison of CO₂ mass partitioning from dual-porosity model and dual-permeability model after 5 years of continuous CO₂ injection. Total injected CO₂ mass is 2.7988×10^5 kg per meter in the y direction.

Table 4.1 shows the CO₂ mass partitioning between fracture domain and rock matrix domain. Both models predict that a larger amount of CO₂ mass is stored in the rock matrix than in the fractures. The dual-permeability model shows 2% more CO₂ mass in the rock matrix than the dual-porosity model. This is consistent with the observation from Figure 4.2 that CO₂ saturation is higher in the plume of the rock matrix domain from the dual-permeability model.

When permeability of the rock matrix increases from 1 mD to 10 mD while permeability of the fractures remains at 1 D, the dual-porosity model reports 80.57% CO₂ mass in the rock matrix and the dual-permeability model reports 81.49% CO₂ mass in the rock matrix. The difference is 0.92%, which is smaller than the 2% difference from the original case. As expected, both models place more CO₂ in the rock matrix when matrix permeability increases because their mass transfer functions, equations (33) and (36), contain the value of matrix permeability but not that of the fracture permeability. Further investigation is needed and other aspects of the model outputs such as the saturation

variability in the plumes should be compared to explain why the percentage difference of the two models is reduced.

Discussion above indicates that the dual-porosity model, a simpler model, provides similar results than the dual-permeability model. However, we are going to focus on the dual-permeability models in the sections below because dual-permeability models are more general.

4.2 Dual-Permeability: Comparison between VE-3D Model and Full-3D Model

4.2.1 Different Simulation Times

In this section I present results from the VE-3D model after 1 year and 5 years of continuous CO₂ injection, and compare them with the results from two versions of the Full-3D model: “Full-3D BC model”, which uses Brooks-Corey relative permeability curve in the fracture domain, and “Full-3D linear model”, which uses linear relative permeability curve in the fracture domain. Both versions use Brooks-Corey relative permeability curve in the rock matrix domain. Full-3D BC model is the default version of Full-3D model. Full-3D linear model is created solely to test if VE-3D code has bugs and is presented and discussed only in this section.

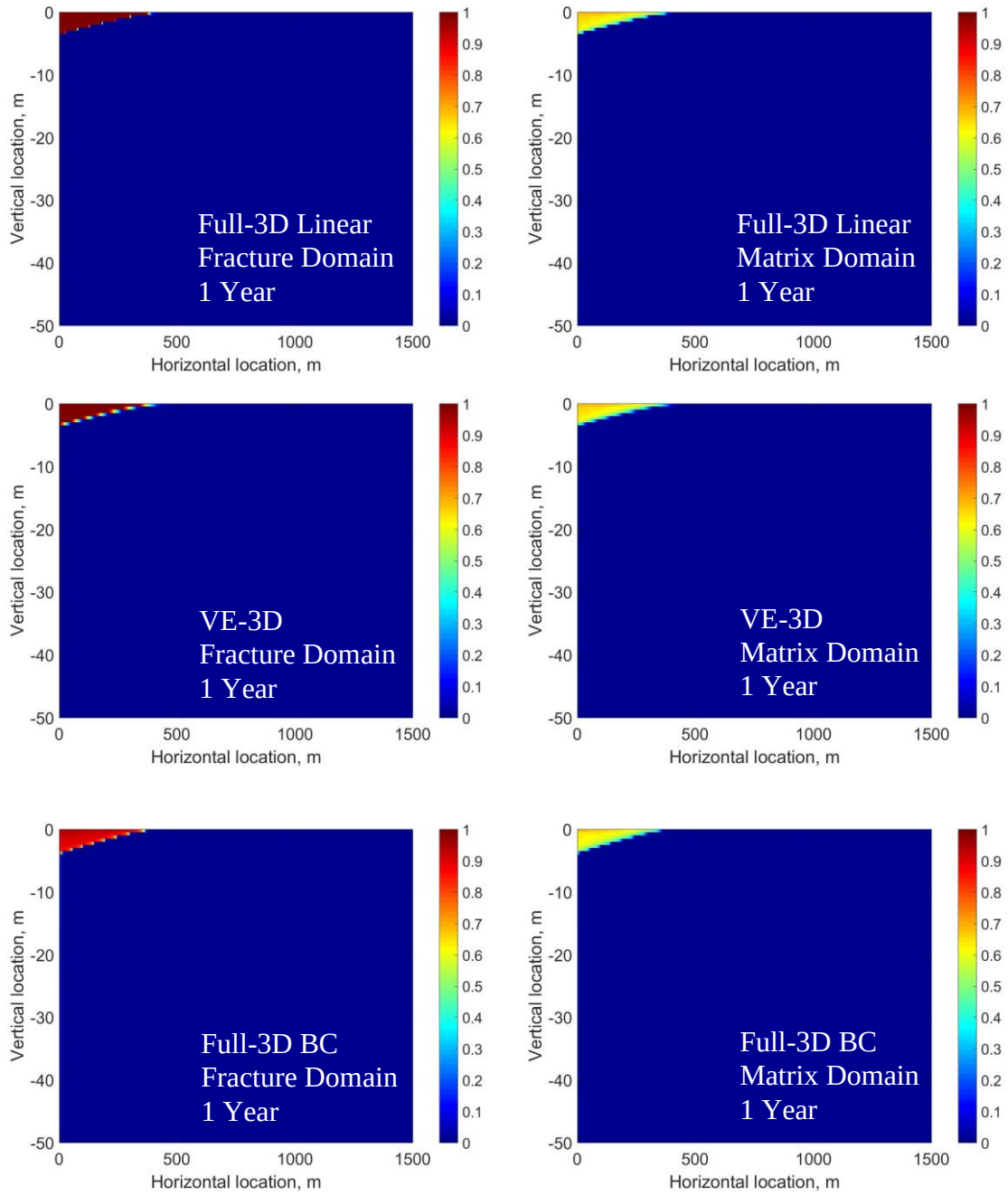


Figure 4.3 Comparison of CO₂ saturation graphs from Full-3D linear model (top), VE-3D model (middle), and Full-3D BC model (bottom) after 1 year of continuous CO₂ injection.

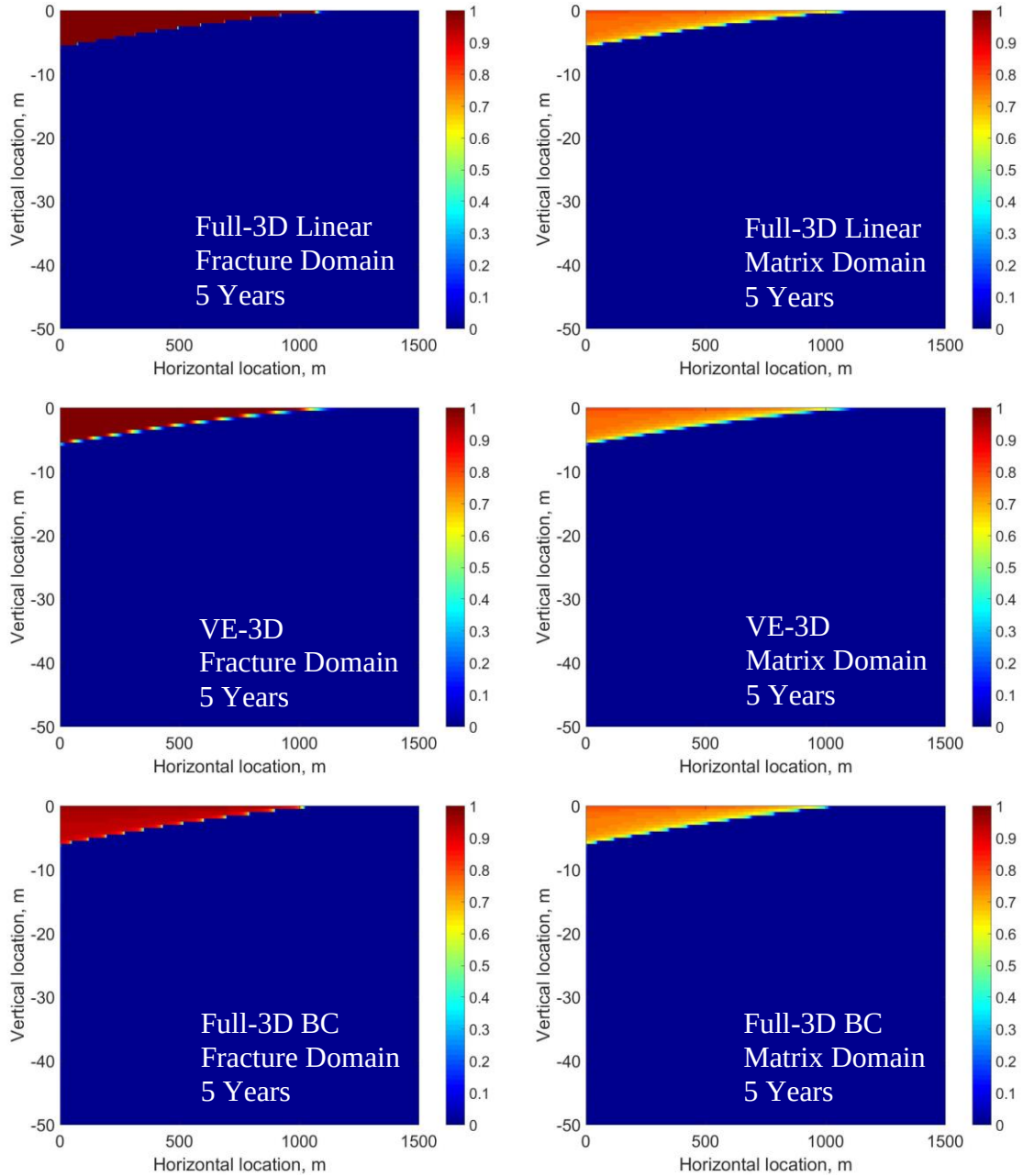


Figure 4.4 Comparison of CO₂ saturation graphs from Full-3D linear model (top), VE-3D model (middle), and Full-3D BC model (bottom) after 5 years of continuous CO₂ injection.

As expected, saturation and plume migration distance increase with time. The shapes of CO₂ plumes are similar for the three models. VE-3D model shows slightly longer leading edges and thinner plume than both versions of Full-3D model due to the sharp

interface assumption. In a 3D model, the Brooks-Corey curve suggests that the relative permeability would decrease with the saturation. When CO₂ saturation in a grid cell is low, CO₂ in that cell cannot move further to the next cell. However, in a vertically-integrated model such as the VE model, relative permeability is one at any point within the CO₂ plume. As a result, some CO₂ can always move further to the next cell and this creates long leading edges. In the 5-year simulation results, the length of the leading edges of CO₂ plume is 1160 meters in VE-3D model and 1020 meters in Full-3D models. Next we investigate the vertical pressure profiles in the three models after 1 year and 5 years of CO₂ injection.

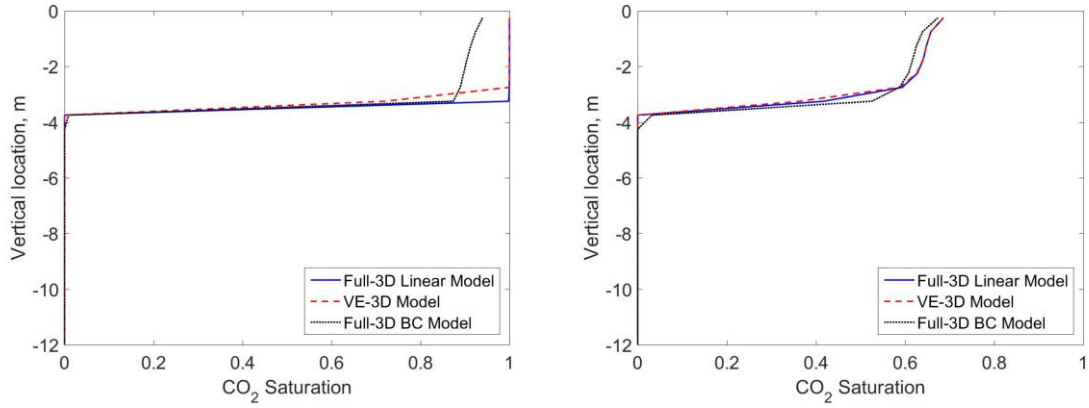


Figure 4.5 Comparison of vertical CO₂ saturation profiles 20 meters from the injection well in the fracture domain (left) and the rock matrix domain (right) after 1 year of continuous CO₂ injection. Only showing top 12 meters of the domain.

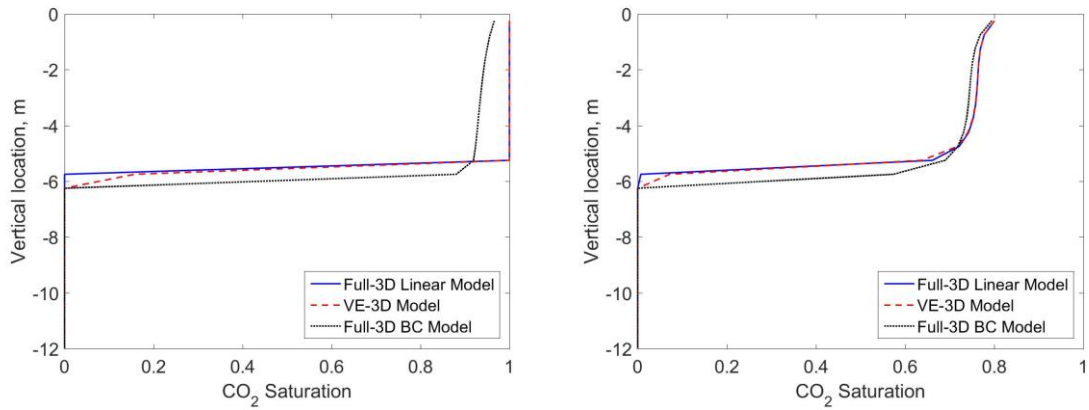


Figure 4.6 Comparison of vertical CO₂ saturation profiles 20 meters from the injection well in the fracture domain (left) and the rock matrix domain (right) after 5 years of continuous CO₂ injection. Only showing top 12 meters of the domain.

Figure 4.6 shows vertical saturation profiles of CO₂ 20 meters from the injection well after 5 years of injection. CO₂ saturation profiles within the plume are similar for the three models. Specifically, CO₂ saturation is 1 throughout the plume depth in the fracture domains of VE-3D model and Full-3D linear model due to the complete segregation of CO₂ and brine. CO₂ saturation varies from 0.95 to 0.9 in the fracture domain of Full-3D

BC model because the vertical segregation is not completed and there is some brine in the CO₂ plume. In the rock matrix domain, CO₂ saturations vary similarly in the CO₂ plume from 0.8 at the top to 0.6 at the bottom of the plume. Slightly lower CO₂ saturation in the fractures of the Full-3D BC model pushes less CO₂ to the rock matrix through the mass exchange flux, leading to slightly lower CO₂ saturation in the rock matrix domain in the Full-3D BC model. Due to the smaller CO₂ saturation in the plume, Full-3D BC model reports slight larger plume depth. CO₂ saturation equals to zero below the CO₂-brine interface in all three models.

	Model	Percentage of Injected CO ₂ Mass in the Fracture Domain	Percentage of Injected CO ₂ Mass in the Rock Matrix Domain
1 year, total injected CO ₂ mass = 5.5976×10^4 kg/m	Full-3D Linear Model	25.62%	74.38%
	VE-3D Model	25.54%	74.46%
	Full-3D BC Model	24.95%	75.05%
5 years, total injected CO ₂ mass = 2.7988×10^5 kg/m	Full-3D Linear Model	21.83%	78.17%
	VE-3D Model	21.74%	78.26%
	Full-3D BC Model	21.09%	78.91%

Table 4.2 Comparison of CO₂ mass partitioning from VE-3D model and Full-3D models after 1 year (in blue) and 5 years (in orange) of continuous CO₂ injection.

Last but not least, we look at the CO₂ mass partitioning in the fracture domain and the rock matrix domain from the three models after 1 year and 5 years of CO₂ injection. All three models indicate that more CO₂ is stored in the rock matrix than in the fractures and the percentage of injected CO₂ that is stored in the rock matrix increases with time. The Full-3D BC model predicts the most mass in rock matrix, followed by the VE-3D

model and the Full-3D linear model in both 1-year and 5-year cases. The difference of mass partitioning between VE-3D model and Full-3D linear model is smaller than the difference between VE-3D model and Full-3D BC model.

The similarity between results from VE-3D model and from Full-3D linear model suggests that the VE-3D model is working properly. For the Full-3D model, using a linear relative permeability curve in the fracture domain means the segregation occurs faster as the relative permeability curve of the brine does not go to near zero for low brine saturation. As a result, no brine is trapped in the plume. When segregation occurs faster, results from the VE-3D and Full-3D models are supposed to be very similar and they actually do, as seen from simulation results above.

Since VE-3D model is working, it can be compared to the Full-3D BC model, which uses the Brooks-Corey relative permeability curve in the fractures and is the default setting of a Full-3D model. Table 4.2 indicates that the difference between the percentage CO₂ mass in the rock matrix domains of the VE-3D model and the Full-3D BC model is 0.59% in the one-year case and 0.65% in the 5-year case. This result suggests that the VE-3D model and the Full-3D BC model do not get closer as simulation time increases, which contradicts the understanding from the single domain model that the VE and 3D models would become closer as time goes on. The mass transfer functions may have brought additional complexities that change the dynamics of the dual-continuum system. As a result, we do not get the same behavior.

We also plot fluid pressure versus “distance from the injection well” at a given depth to analyze pressure outputs from the models. Figures 4.7 and 4.8 show horizontal pressure profiles from two versions of Full-3D model and VE-3D model at 0.25 meters,

3.25 meters, 4.25 meters, 9.25 meters, 25.25 meters, and 49.75 meters below the top of the formation after 1 year of injection. Fractures results are shown first, followed by rock matrix results.

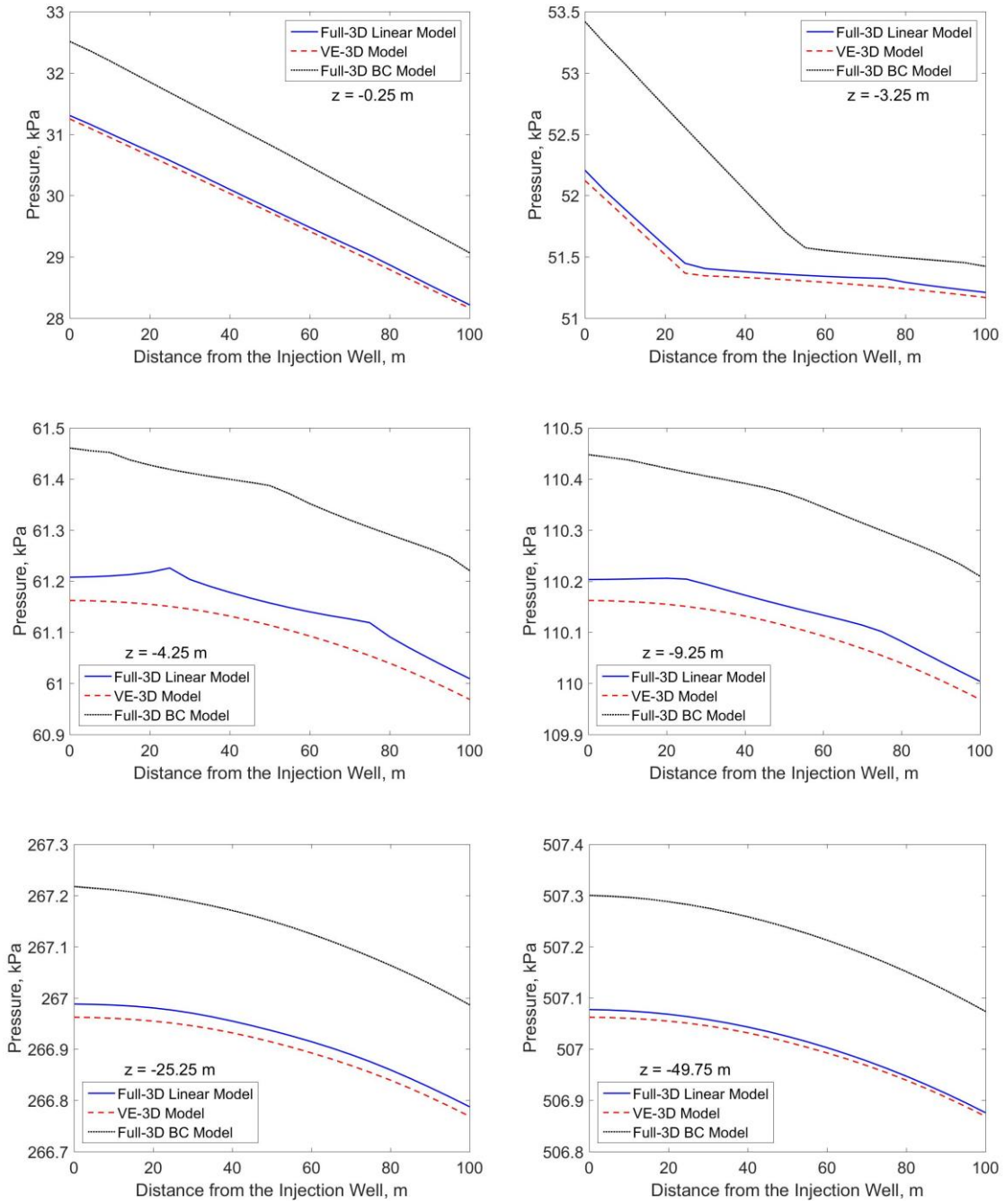


Figure 4.7 Horizontal pressure profiles in the fracture domain at different depths from VE-3D model and Full-3D models after 1 year of CO₂ injection.

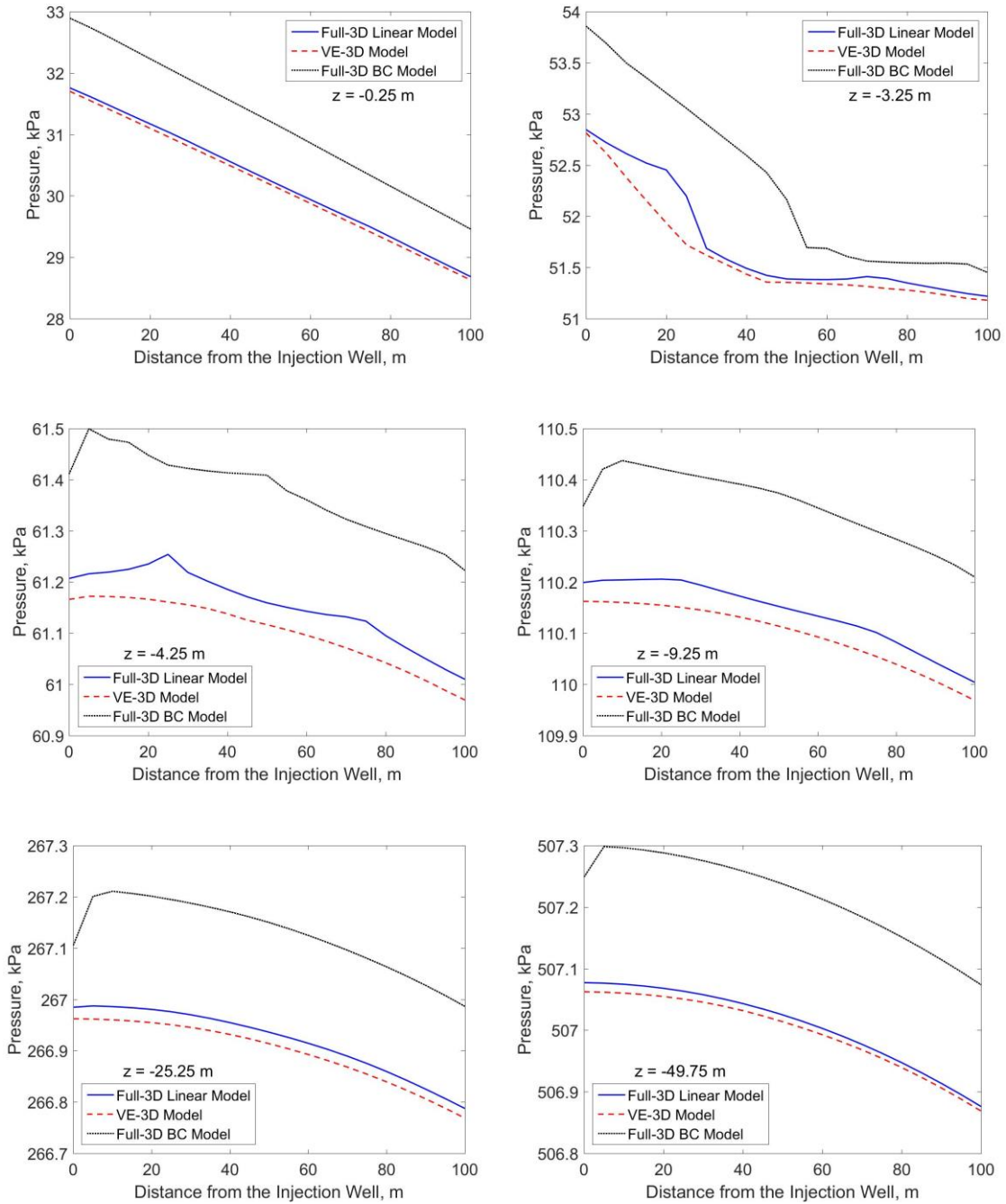


Figure 4.8 Horizontal pressure profiles in the rock matrix domain at different depths from VE-3D model and Full-3D models after 1 year of CO_2 injection.

Figures 4.7 and 4.8 show that pressure at a given depth tends to decrease with increasing distance from the injection well. This is reasonable because CO_2 is pushed into

the geological formation at the injection well so pressure should be high close to the well. Fluid pressure gradually dissipates to the whole geological formation and decreases as we move away from the injection well.

Secondly, pressure from the VE-3D model is always smaller than pressure from the two Full-3D models. The absolute difference is around 50 Pascal when compared to the Full-3D linear model and 300 Pascal when compared to the Full-3D BC model. These differences do not change much with depth. The relative difference is low, ranging from 0.1-1% at the top of the formation to 0.01-0.06% at the bottom of the formation.

Also, in the Full-3D linear model, a strong discontinuity pattern in the slope of the curve, with a wavy shape in the horizontal pressure profile occurs when depth of pressure is close to the CO₂-brine interface. In the Full-3D BC model, a moderate discontinuity pattern occurs around the depth of the interface. The discontinuity patterns disappear when pressure profiles are plotted well above or below the interface. No discontinuity is seen in pressure profiles of the VE-3D model. The wavy pattern is also observed in 3D models for a single domain when permeability is large (e.g. 1000 mD) while it is absent when permeability is low (e.g. 1 mD). The wavy pattern also appears for finer spatial and temporal discretization. Therefore, the patterns are likely to represent physical mechanisms in the geological formation, as opposed to being created purely by the numerical simulation. Modeling results from a single domain model show that as the CO₂ plume front progresses from left to right in the domain, the majority of brine moves to the right, while some brine moves down and a small amount of brine moves to the left. The brine that moves down and moves to the left could disturb the pressure at depths around

the interface and create those wavy horizontal pressure profiles. Further investigation is needed to confirm these hypotheses.

4.2.2 Different Permeability for the Fractures

Iding and Ringrose (2010) estimated that the permeability of fractures ranges from 100 mD to 1D and the permeability of the rock matrix is around 10 mD in the In Salah CO₂ storage site. In this section, we use smaller permeability for the rock matrix domain and look at extreme cases in the fracture domain: permeability of the rock matrix is fixed at 1 mD while permeability of the fractures varies from 10 mD to 100 D. The simulation time is 5 years.

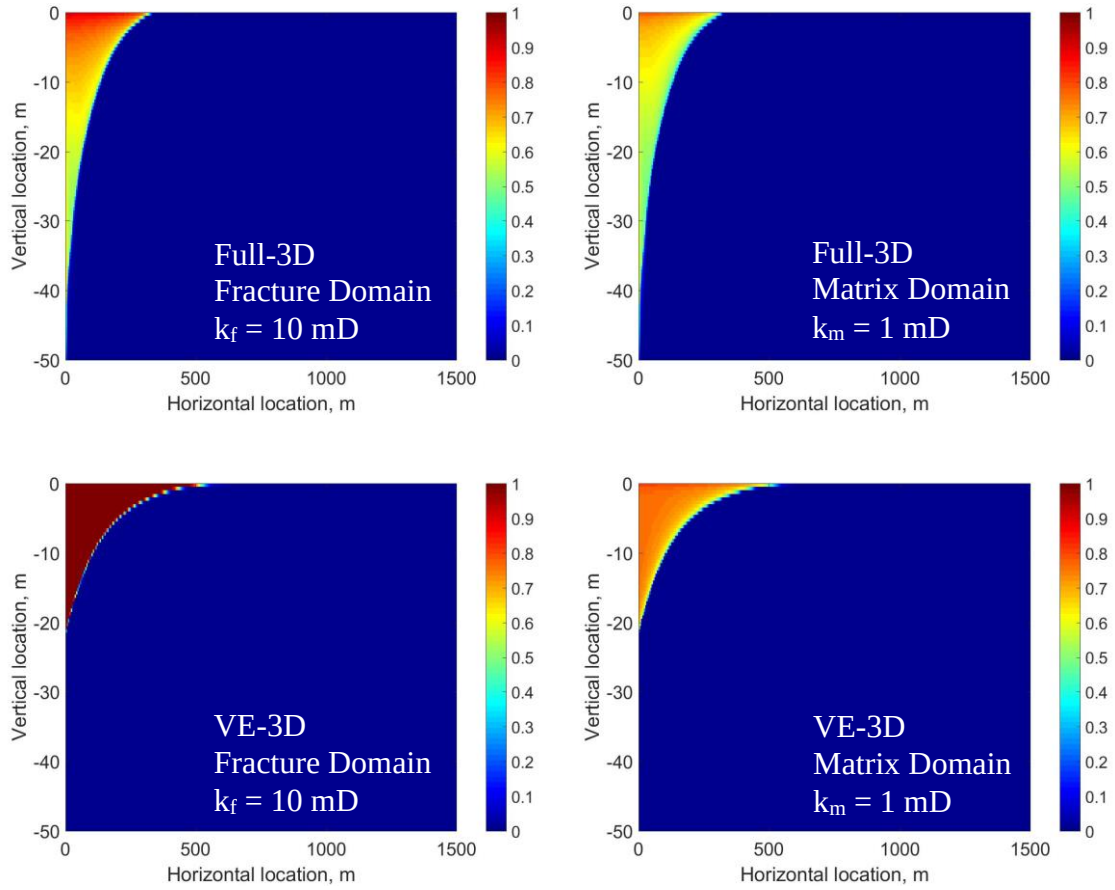


Figure 4.9 Comparison of CO₂ saturation graphs from Full-3D model (top) and VE-3D model (bottom) after 5 years of continuous CO₂ injection when fracture permeability is 10 mD and matrix permeability is 1 mD.

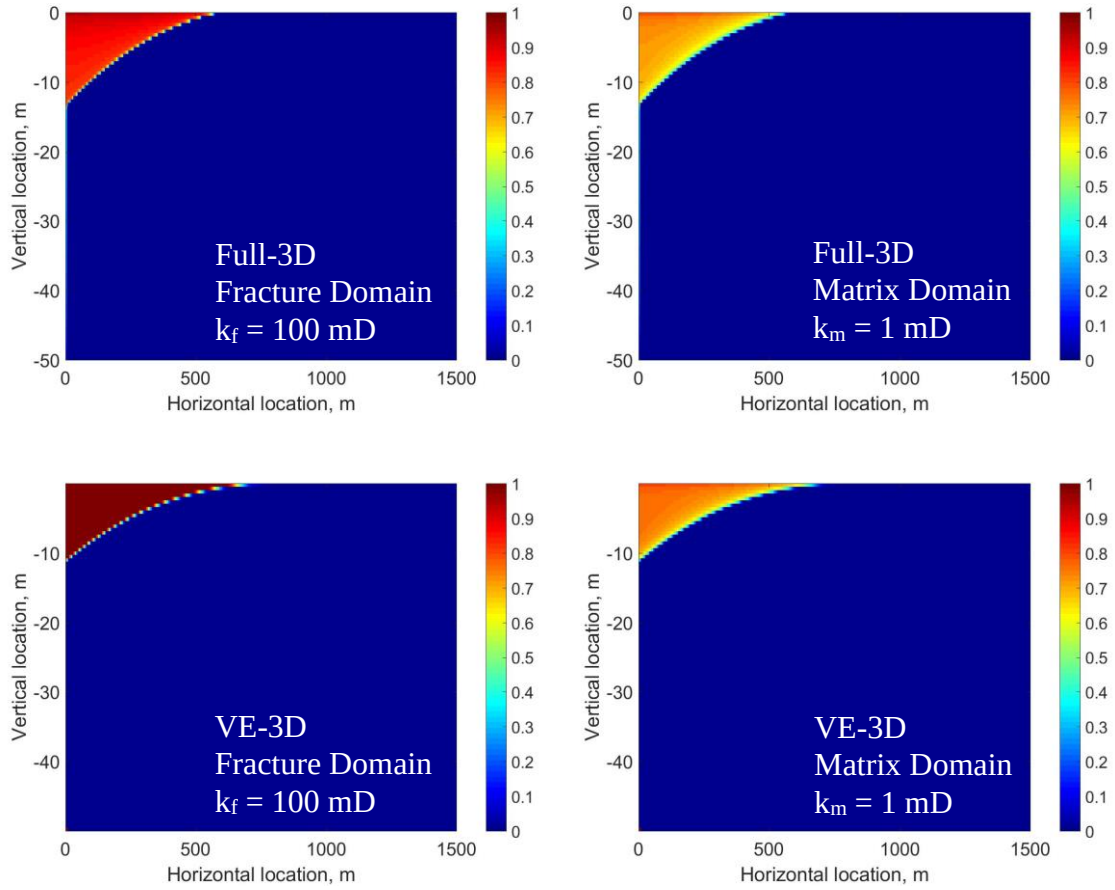


Figure 4.10 Comparison of CO₂ saturation graphs from Full-3D model (top) and VE-3D model (bottom) after 5 years of continuous CO₂ injection when fracture permeability is 100 mD and matrix permeability is 1 mD.

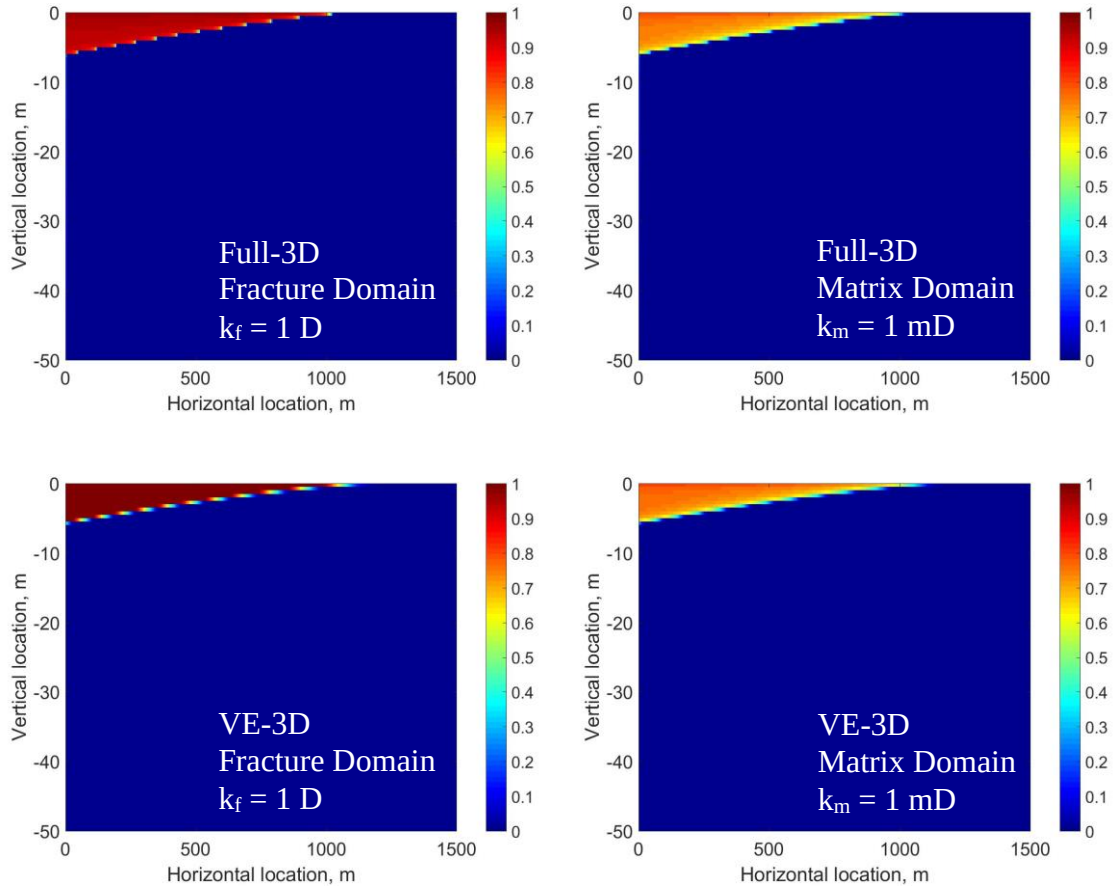


Figure 4.11 Comparison of CO₂ saturation graphs from Full-3D model (top) and VE-3D model (bottom) after 5 years of continuous CO₂ injection when fracture permeability is 1 D and matrix permeability is 1 mD.

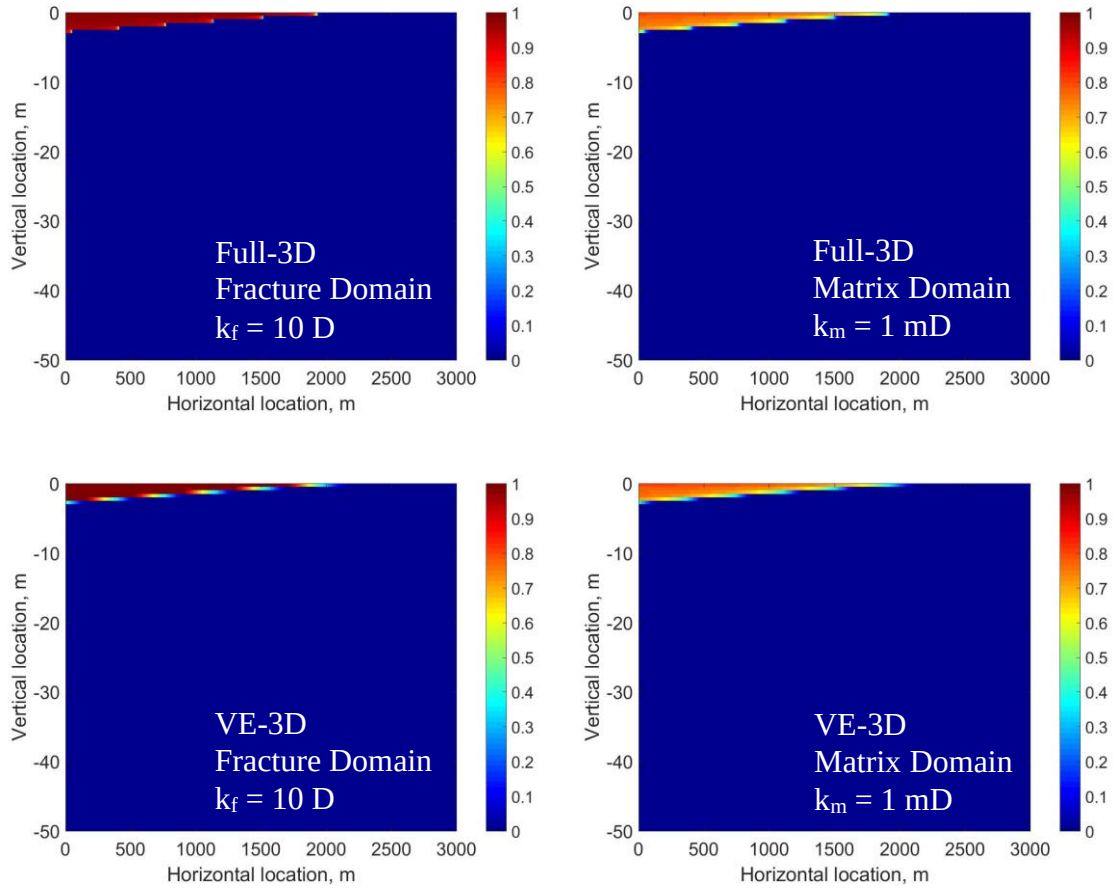


Figure 4.12 Comparison of CO₂ saturation graphs from Full-3D model (top) and VE-3D model (bottom) after 5 years of continuous CO₂ injection when fracture permeability is 10 D and matrix permeability is 1 mD. Notice that the length of the domain changes from 1500 meters to 3000 meters.

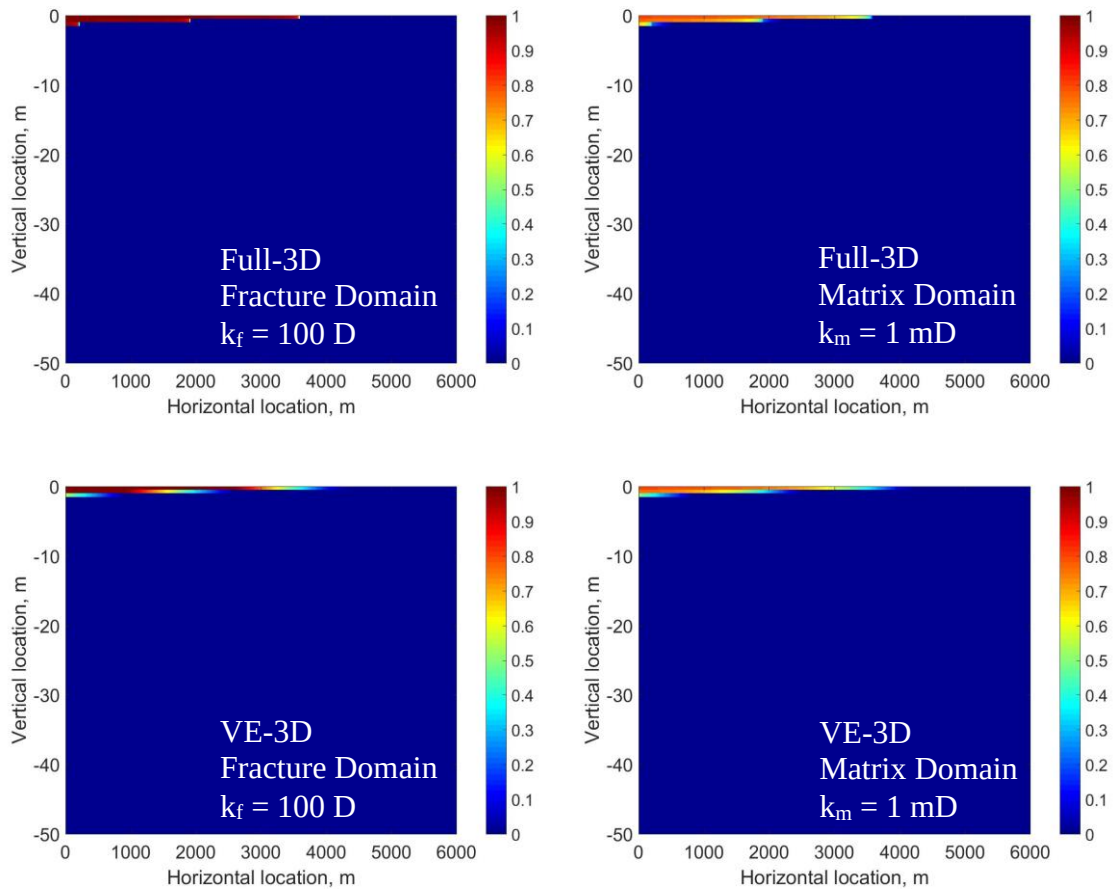


Figure 4.13 Comparison of CO₂ saturation graphs from Full-3D model (top) and VE-3D model (bottom) after 5 years of continuous CO₂ injection when fracture permeability is 100 D and matrix permeability is 1 mD. Notice that the length of the domain changes from 3000 meters to 6000 meters.

Fracture Permeability	Model	Percentage of Injected CO ₂ Mass in the Fracture Domain	Percentage of Injected CO ₂ Mass in the Rock Matrix Domain
0.01 D	Full-3D	18.94%	81.07%
	VE-3D	22.02%	77.98%
0.1 D	Full-3D	20.41%	79.59%
	VE-3D	21.94%	78.06%
1 D	Full-3D	21.09%	78.91%
	VE-3D	21.74%	78.26%
10 D	Full-3D	21.37%	78.63%
	VE-3D	21.34%	78.66%
100 D	Full-3D	21.35%	78.65%
	VE-3D	20.64%	79.36%

Table 4.3 Comparison of CO₂ mass partitioning from VE-3D model and Full-3D model after 5 years of continuous CO₂ injection. Fracture permeability varies from 10 mD to 100 D. Rock matrix permeability is fixed at 1 mD. Total injected CO₂ mass is 2.7988×10^5 kg/m.

Figures 4.9 – 4.13 show that CO₂ plumes from the VE-3D model and Full-3D model become more and more similar when fracture permeability increases from 10 mD to 100 D. The length of the domain is 1500 meters for Figures 4.9 – 4.11, 3000 meters for Figure 4.12, and 6000 meters for Figure 4.13. When fractures have low permeability such as 10 mD, the size of the CO₂ plume and saturation variability in the plume from the two models are very different. However, when fractures have larger permeability such as 1 D, the CO₂ plume from two models are very similar.

Court et al. (2012) states that modeling results from the VE and 3D models match well when permeability is larger than 100 mD. However, Figures 4.9 – 4.13 show that the saturation predictions from the VE-3D and Full-3D models match well when fracture permeability is larger than 1 D. The 100 mD threshold from the single-domain VE model therefore may not hold for the dual-domain VE-3D model. It is possible that CO₂ exchange between the two domains makes the segregation timescale longer because some

CO₂ is sent to the rock matrix very early in time and the effects of those transfers stay in the dual-continuum system from thereon.

From Table 4.3, we see that the mass partitioning results from the two models become closer when fracture permeability increases from 10 mD to 10 D, but diverge as fracture permeability increases from 10 D to 100 D. The divergence may be due to the numerical diffusion in the Full-3D model. A closer look of the top left figure of Figure 4.15 reveals that the CO₂ plume only occupies the top three rows in the domain. When the plume is thin relative to the height of the grid cell, numerical diffusion in the Full-3D model makes the result less accurate. This problem could be solved by using finer discretization of the domain. However, run-time would become very long for the high-permeability high-discretization case.

From Table 4.3 we see that the VE-3D model provides close estimation of CO₂ storage capacity to the Full-3D model when fracture permeability is larger than 1 D, which is a reasonable permeability for well-connected fractures. At the same time, the error for low fracture permeability is relatively small, so the VE-3D model might be acceptable for capacity estimates even when fractures have low permeability. In sum, the VE-3D model is appropriate for both saturation prediction and capacity estimations when fracture permeability is larger than 1 D. For fracture permeability smaller than 1 D, the VE-3D model should not be used to predict saturation but might be acceptable for capacity estimation.

This conclusion is contingent to the 5-year simulation time and the conclusion might change when we look at longer time horizon such as 50 years. As discussed in Section

4.2.1, the complexities added to the dual-continuum system through mass transfer functions could change the long-term results.

The computational advantage of the VE-3D model shows up in the test cases in this section. When fracture permeability is high, the Full-3D model requires small time steps to capture the vertical migration of CO₂, leading to long run times. However, the time steps of the VE-3D model are less dependent on the fracture permeability. In addition to allowing larger time steps, the VE-3D model runs faster in each step than the Full-3D model due to the reduction in dimensionality. The combination of these two advantages suggests significantly shorter run time.

4.3 Dual-Permeability: VE-3D Model with Different Forms of Mass Transfer Functions

In this section, the impacts of the different driving forces of mass transfer are investigated based on results from the VE-3D model. We present three cases in which different types of mass transfer functions are used. As discussed in Section 2.4.1, the mass transfer functions for dual-permeability models take into account three driving forces: viscous force, capillary force, and gravitational force. Since we neglect capillary pressure in these test cases, the mass transfer function for each phase only has a pressure component, which is the difference between fluid pressure in fractures and rock matrix, to account for the viscous force and a saturation component, which is the difference between CO₂ saturation in the fractures and the rock matrix, to account for the gravitational force. We run three test cases. In the first case, mass transfer functions have pressure and saturation

components. In the second case, mass transfer functions only have pressure components. In the third case, mass transfer functions only have saturation components. The simulation time is 5 years in all three cases and CO₂ is injected only into the fracture domain.

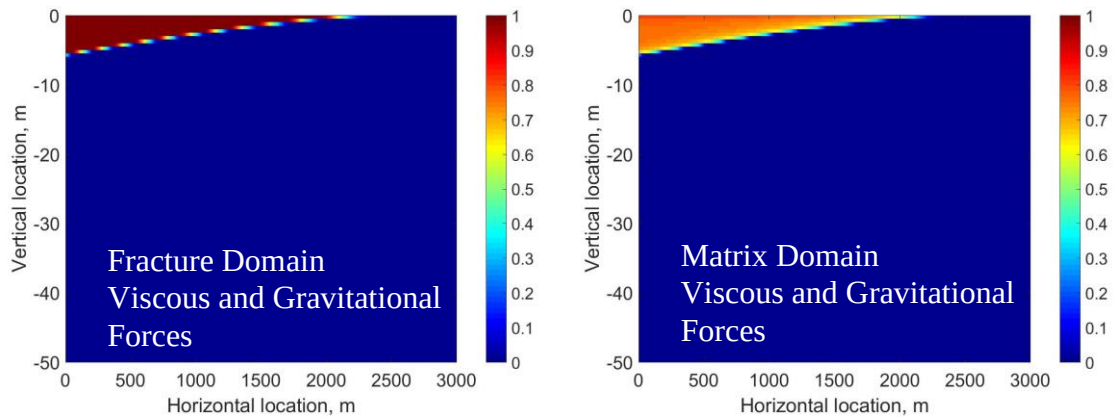


Figure 4.14 CO₂ saturation graphs from VE-3D model after 5 years of continuous CO₂ injection when mass transfer functions of CO₂ and brine have pressure components and saturation components.

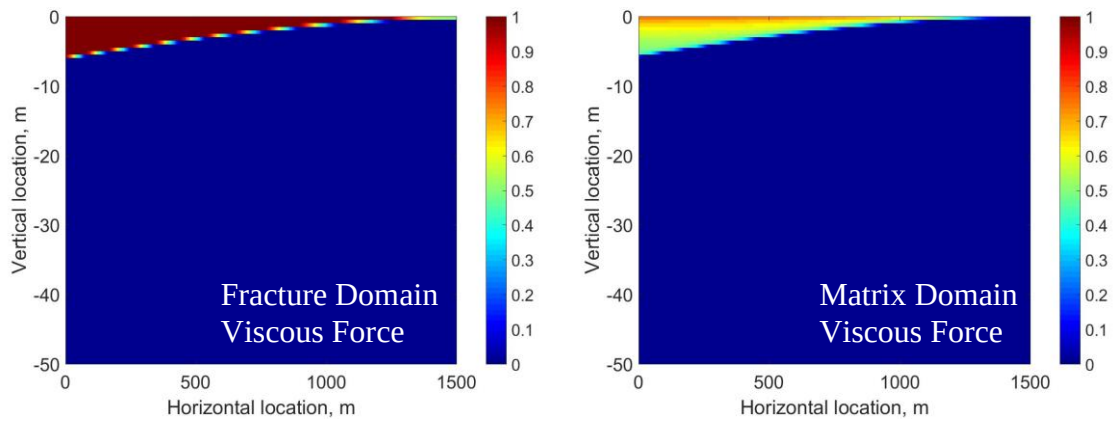


Figure 4.15 CO₂ saturation graphs from VE-3D model after 5 years of continuous CO₂ injection when mass transfer functions of CO₂ and brine only have pressure components.

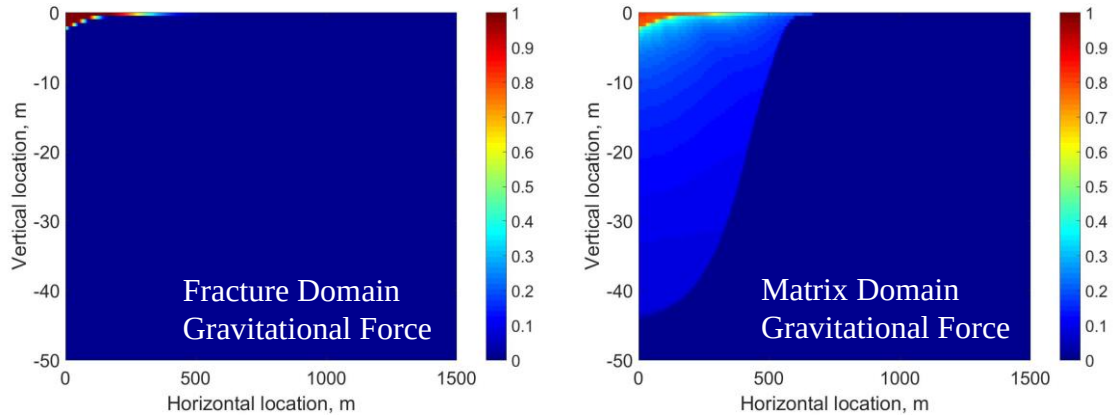


Figure 4.16 CO₂ saturation graphs from VE-3D model after 5 years of continuous CO₂ injection when mass transfer functions of CO₂ and brine only have saturation components.

	Percentage of Injected CO ₂ Mass in the Fracture Domain	Percentage of Injected CO ₂ Mass in the Rock Matrix Domain
Pressure and saturation components	21.74%	78.26%
Pressure component only	27.54%	71.08%
Saturation component only	2.07%	97.93%

Table 4.4 Comparison of CO₂ mass partitioning from VE-3D model in three cases. In the second case, percentages do not sum to 100% because the plumes have long leading edges and 1.38% of CO₂ runs out of the dual-continuum system through the right boundary. Total injected CO₂ mass is 2.7988×10^5 kg/m for all three cases.

Comparing the case that only considers the pressure component (Figure 4.15) to the case with both components (Figure 4.14), we see that when mass transfer functions only have pressure components, the plume in the fractures is thinner and longer, the plume in the rock matrix has smaller CO₂ saturation, and there is less CO₂ mass in the rock matrix. CO₂ is also more concentrated at the top of the plume. Since CO₂ is injected into the

fracture domain, these observations indicate that the viscous force itself transfers CO₂ from fractures to rock matrix but the transfer is smaller compared to the first case in which both viscous force and gravitational force are present.

Comparing the case that only considers the saturation component (Figure 4.16) to the case with both components (Figure 4.14), we see that when mass transfer functions only have saturation components, the plume in the fractures is much smaller, CO₂ exists in a much larger space in the rock matrix, and the majority of the CO₂ mass is in the rock matrix. There are two implications. Firstly, the gravitational force is very strong in transferring CO₂ from fractures to the rock matrix. Secondly, viscous forces act in the opposite direction to the gravitational forces, moving some CO₂ mass from the rock matrix back into the fractures. Without viscous forces, the majority of CO₂ is transferred to the rock matrix soon after it is injected to the fracture and stays in the rock matrix. CO₂ flows slowly in the less permeable rock matrix and thus forms the very large CO₂ plume in the rock matrix domain.

In sum, gravitational forces bring CO₂ from fractures into the rock matrix as long as there exists a saturation difference between the two domains. However, viscous forces act dynamically and change the direction of CO₂ mass transfer depending on the strength of the gravitational forces. When gravitational forces are strong, viscous forces act in the opposite direction to obstruct CO₂ flux from the fractures to the rock matrix, as seen from the comparison between the case that only considers the pressure component and the case that considers both components. When the gravitational forces are too weak or absent, the viscous forces bring CO₂ from the fractures into the rock matrix, as seen from the

comparison between the case that only considers the saturation component and the case that considers both components.

Chapter 5

Conclusions and Future Work

In this thesis, a dual-continuum model called the VE-3D model is developed and compared to the Full-3D model to study CO₂ injection and migration in fractured geological formations. The VE-3D model uses a VE model in the fracture domain and a 3D model in the rock matrix domain. VE models solve the vertically-integrated 3D governing equations by assuming rapid and complete buoyant segregation of the two fluid phases. Comparing to 3D models, VE models are computationally less costly but only suitable for formations with relatively large permeability. VE and 3D models have been used widely to study different aspects of CO₂ injection and migration. It is the coupling between the VE model (in fractures) and the 3D model (in matrix) that makes the VE-3D model an innovative creation.

Comparing to the Full-3D model, which is the conventional approach for modeling CO₂ injection in fractured geological formations, the VE-3D model vertically integrates the mass transfer functions and reconstructs the pressure and saturation in the fracture domain in each time step. These operations are much simpler than solving for all pressure and saturation over the fracture domain and they give the VE-3D model computational advantages over the Full-3D model. The mass transfer functions represent fluid mass

transfer between the fracture and rock matrix domains. Mass transfer functions in this thesis are adopted from petroleum engineering and have different forms for the dual-permeability and dual-porosity models. The test cases in this thesis are based on a typical geological formation and are slices in the (x,z) plane. The domain has a length of at least 1500 meters and a height of 50 meters, and is discretized by at least 30000 grid cells.

Results from the VE-3D model and the Full-3D model are compared in four aspects: shape of the CO₂ plume, saturation of CO₂ in the domain, pressure of CO₂ in the domain, and CO₂ mass partitioning in the fracture and rock matrix domains. The VE-3D model provides results comparable to the Full-3D model and runs significantly faster. In the test cases, 1.4×10^5 tons of CO₂ is injected and stored in the fractured reservoir during the course of 5 years. We also qualitatively determine that the VE model is most suitable for fractures whose permeability is higher than 1 D when we look at the 5-year horizon. Last but not least, results from the VE-3D model with different forms of mass transfer functions are examined and they provide us insight into the two-phase dual-continuum system. Gravitational forces play an important role driving fluid mass transfer between the two domains. Viscous forces enhance or obstruct the mass transfer depending on the strength of gravitational forces and viscous forces would not push much CO₂ into the rock matrix domain on their own.

There are several directions in which the new modeling approach can be extended in the future. For example, different modeling options are available for rock matrix in addition to the 3D model, which is used in both the VE-3D model and the Full-3D model. One option is to use the Dynamic Reconstruction (DR) model in the rock matrix domain (Guo *et al.*, 2014). The DR modeling approach uses the vertically-integrated framework

while relaxing the vertical equilibrium assumption and thus captures the vertical dynamics of CO₂ and brine. It is also much more computationally efficient than the 3D model due to the reduced dimensionality. We can replace the 3D model with the DR model to make the dual-continuum models run even faster. This group of models with different assumptions and different levels of complexities would provide us useful and effective tools to analyze CO₂ migration in different types of fractured geological formations.

Another logical next step is to test the two models with datasets from real injection sites. The comparison between model output and real monitoring data would allow us to identify the advantages and limitations of these modeling approaches and provide us insight on how to improve the models for specific CO₂ injection sites.

The models described in this thesis can provide accurate and efficient estimation of underground CO₂ storage capacity. This would allow policymakers to make informed decisions when deciding on carbon mitigation strategies. Also, our models predict how far CO₂ plumes migrate and how fluid pressure increases in the geological formation. This information would allow us to take precautionary measures to minimize the risk of CO₂ leakage. For example, abandoned hydrocarbon wells in the range of the expected CO₂ front should be plugged and carefully monitored. Overall, our models would enhance our understanding of geological storage of CO₂. Timely development and safe deployment of CCS would build human beings a bridge to the better and sustainable future.

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Figure 1.1 – the figure is modified from Carbon Science & Engineering Research Group, University of Utah, produced by Genevieve Young, [Online; accessed 2017-04-10; <https://co2.egi.utah.edu/>].

Figure 1.2 – the figure is taken from Department of Energy National Energy Technology Laboratory (2010), Carbon sequestration atlas of the United States and Canada, third edition.

Figure 1.3 – the figure is modified from Bo Guo’s final public oral exam slides, Subsurface Hydrology Research Group, Princeton University, May 2016.

Figure 2.1 – the figures are taken from Bo Guo’s presentation “vertically integrated dual continuum methods for CO₂ migration in fractured saline aquifers” at Berkeley Lawrence National Laboratory, December 2016.

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Appendix

In this Appendix, I would like to show how I derive the discrete governing equations for Full-3D model and VE-3D model using the finite volume method.

Appendix A: Discretized Governing Equations for Full-3D

Model

In the fracture domain, the pressure equation is equation (30) and is repeated below.

$$s_c^f c_c^f \frac{\partial p_c^f}{\partial t} + s_b^f c_b^f \frac{\partial p_b^f}{\partial t} + \nabla \cdot \mathbf{u}_c^f + \nabla \cdot \mathbf{u}_b^f = \psi_c^f + \psi_b^f - q_c^{f-m} - q_b^{f-m} \quad (30)$$

Assuming no compressibility and only focusing on the (x,z) plane, the pressure equation becomes

$$\begin{aligned} \frac{\partial}{\partial x} \left(-\frac{k_{r,c}^f k_x^f}{\mu_c} \frac{\partial p_c^f}{\partial x} \right) + \frac{\partial}{\partial z} \left[-\frac{k_{r,c}^f k_z^f}{\mu_c} \frac{\partial}{\partial z} (p_c^f + \rho_c g z) \right] + \frac{\partial}{\partial x} \left(-\frac{k_{r,b}^f k_x^f}{\mu_b} \frac{\partial p_b^f}{\partial x} \right) \\ + \frac{\partial}{\partial z} \left[-\frac{k_{r,b}^f k_z^f}{\mu_b} \frac{\partial}{\partial z} (p_b^f + \rho_b g z) \right] = \psi_c^f + \psi_b^f - q_c^{f-m} - q_b^{f-m} \end{aligned} \quad (45)$$

All variables in equations (46) – (50) are from the fracture domain. Their superscripts f are omitted. Let $p_{b,i,j}$ denotes p_b in the grid cell that is characterized by i and j . i indicates the horizontal location of a grid cell and j indicates the vertical location. $p_{b,i-1,j}$ represents p_b in the grid cell to the left of $p_{b,i,j}$'s grid cell and $p_{b,i,j+1}$ represents p_b in the grid cell on the top of $p_{b,i,j}$'s grid cell. The pressure equation in the grid cell (i,j) becomes

$$\begin{aligned}
& \frac{\frac{k_{r,c}k_x}{\mu_c} \frac{\partial p_b}{\partial x} \Big|_{i+\frac{1}{2},j} - \frac{k_{r,c}k_x}{\mu_c} \frac{\partial p_b}{\partial x} \Big|_{i-\frac{1}{2},j}}{\delta x_i} + \frac{\frac{k_{r,c}k_x}{\mu_c} \frac{\partial p^{cap}}{\partial x} \Big|_{i+\frac{1}{2},j} - \frac{k_{r,c}k_x}{\mu_c} \frac{\partial p^{cap}}{\partial x} \Big|_{i-\frac{1}{2},j}}{\delta x_i} \\
& + \frac{\frac{k_{r,c}k_z}{\mu_c} \frac{\partial p_b}{\partial z} \Big|_{i,j+\frac{1}{2}} - \frac{k_{r,c}k_z}{\mu_c} \frac{\partial p_b}{\partial z} \Big|_{i,j-\frac{1}{2}}}{\delta z_j} \\
& + \frac{\frac{k_{r,c}k_z}{\mu_c} \frac{\partial p^{cap}}{\partial z} \Big|_{i,j+\frac{1}{2}} - \frac{k_{r,c}k_z}{\mu_c} \frac{\partial p^{cap}}{\partial z} \Big|_{i,j-\frac{1}{2}}}{\delta z_j} \\
& + \frac{\frac{k_{r,c}k_z}{\mu_c} \rho_c g \Big|_{i,j+\frac{1}{2}} - \frac{k_{r,c}k_z}{\mu_c} \rho_c g \Big|_{i,j-\frac{1}{2}}}{\delta z_j} \\
& + \frac{\frac{k_{r,b}k_x}{\mu_c} \frac{\partial p_b}{\partial x} \Big|_{i+\frac{1}{2},j} - \frac{k_{r,b}k_x}{\mu_c} \frac{\partial p_b}{\partial x} \Big|_{i-\frac{1}{2},j}}{\delta x_i} \\
& + \frac{\frac{k_{r,b}k_z}{\mu_c} \frac{\partial p_b}{\partial z} \Big|_{i,j+\frac{1}{2}} - \frac{k_{r,b}k_z}{\mu_c} \frac{\partial p_b}{\partial z} \Big|_{i,j-\frac{1}{2}}}{\delta z_j} \\
& + \frac{\frac{k_{r,b}k_z}{\mu_c} \rho_b g \Big|_{i,j+\frac{1}{2}} - \frac{k_{r,b}k_z}{\mu_c} \rho_b g \Big|_{i,j-\frac{1}{2}}}{\delta z_j} \\
& = - \left(\psi_{c,i,j} + \psi_{b,i,j} - q_c^{f-m}{}_{i,j} - q_b^{f-m}{}_{i,j} \right)
\end{aligned} \tag{46}$$

Multiply both sides of the equation by $\delta x_i \delta z_j$. The first term on the left hand side of equation (46) can be expanded and is shown below

$$\begin{aligned}
& \left(\frac{2}{\mu_c}\right) (k_{r,c_{i+1,j}}) \left(\frac{1}{\frac{1}{k_{x_{i,j}}} + \frac{1}{k_{x_{i+1,j}}}}\right) \left(\frac{\delta z_j}{\Delta x_i}\right) p_{b_{i+1,j}} \\
& - \left[\left(\frac{2}{\mu_c}\right) (k_{r,c_{i+1,j}}) \left(\frac{1}{\frac{1}{k_{x_{i,j}}} + \frac{1}{k_{x_{i+1,j}}}}\right) \left(\frac{\delta z_j}{\Delta x_i}\right) \right. \\
& \left. + \left(\frac{2}{\mu_c}\right) (k_{r,c_{i-1,j}}) \left(\frac{1}{\frac{1}{k_{x_{i-1,j}}} + \frac{1}{k_{x_{i,j}}}}\right) \left(\frac{\delta z_j}{\nabla x_i}\right) \right] p_{b_{i,j}} \\
& + \left(\frac{2}{\mu_c}\right) (k_{r,c_{i-1,j}}) \left(\frac{1}{\frac{1}{k_{x_{i-1,j}}} + \frac{1}{k_{x_{i,j}}}}\right) \left(\frac{\delta z_j}{\nabla x_i}\right) p_{b_{i-1,j}} + \text{second term} \\
& + \dots = - \left(\psi_{c_{i,j}} + \psi_{b_{i,j}} - q_c^{f-m}{}_{i,j} - q_b^{f-m}{}_{i,j} \right) (\delta x_i \delta z_j)
\end{aligned} \tag{47}$$

Other terms in the left hand side of equation (46) can be expanded in the similar way. The transport equation in the fracture domain is equation (31) and is repeated below

$$\phi^f \frac{\partial s_c^f}{\partial t} + s_c^f c_c^f \frac{\partial p_c^f}{\partial t} + \nabla \cdot \mathbf{u}_c^f = \psi_c^f - q_c^{f-m} \tag{31}$$

Assuming no compressibility and only focusing on the (x,z) plane, the transport equation becomes (48) and is discretized into equation (49)

$$\phi \frac{\partial s_c}{\partial t} + \frac{\partial}{\partial x} \left(-\frac{k_{r,c} k_x}{\mu_c} \frac{\partial p_c}{\partial x} \right) + \frac{\partial}{\partial z} \left[-\frac{k_{r,c} k_z}{\mu_c} \frac{\partial}{\partial z} (p_c + \rho_c g z) \right] = \psi_c - q_c^{f-m} \tag{48}$$

$$\begin{aligned}
& \phi \frac{\partial s_{c_{i,j}}}{\partial t} + \frac{\frac{k_{r,c} k_x}{\mu_c} \frac{\partial p_c}{\partial x} \Big|_{i+\frac{1}{2},j} - \frac{k_{r,c} k_x}{\mu_c} \frac{\partial p_c}{\partial x} \Big|_{i-\frac{1}{2},j}}{\delta x_i} + \frac{\frac{k_{r,c} k_z}{\mu_c} \frac{\partial p_b}{\partial z} \Big|_{i,j+\frac{1}{2}} - \frac{k_{r,c} k_z}{\mu_c} \frac{\partial p_b}{\partial z} \Big|_{i,j-\frac{1}{2}}}{\delta z_j} \\
& + \frac{\frac{k_{r,c} k_z}{\mu_c} \rho_c g \Big|_{i,j+\frac{1}{2}} - \frac{k_{r,c} k_z}{\mu_c} \rho_c g \Big|_{i,j-\frac{1}{2}}}{\delta z_j} = - \left(\psi_{c_{i,j}} - q_c^{f-m}{}_{i,j} \right)
\end{aligned} \tag{49}$$

Multiply both sides of the equation by $\delta x_i \delta z_j$. Terms on the left hand side of equation (49) can be expanded. The first and second terms are expanded as an example and are shown below

$$\begin{aligned}
& \phi \frac{s_{c,i,j}^{n+1} - s_{c,i,j}^n}{\Delta t} (\delta x_i \delta z_j) + \left(\frac{2}{\mu_c} \right) (k_{r,c,i+1,j}) \left(\frac{1}{\frac{1}{k_{x,i,j}} + \frac{1}{k_{x,i+1,j}}} \right) \left(\frac{\delta z_j}{\Delta x_i} \right) p_{c,i+1,j} \\
& - \left[\left(\frac{2}{\mu_c} \right) (k_{r,c,i+1,j}) \left(\frac{1}{\frac{1}{k_{x,i,j}} + \frac{1}{k_{x,i+1,j}}} \right) \left(\frac{\delta z_j}{\Delta x_i} \right) \right. \\
& \left. + \left(\frac{2}{\mu_c} \right) (k_{r,c,i-1,j}) \left(\frac{1}{\frac{1}{k_{x,i-1,j}} + \frac{1}{k_{x,i,j}}} \right) \left(\frac{\delta z_j}{\Delta x_i} \right) \right] p_{c,i,j} \\
& + \left(\frac{2}{\mu_c} \right) (k_{r,c,i-1,j}) \left(\frac{1}{\frac{1}{k_{x,i-1,j}} + \frac{1}{k_{x,i,j}}} \right) \left(\frac{\delta z_j}{\Delta x_i} \right) p_{c,i-1,j} + \text{third term} + \dots \\
& = -(\psi_c - q_c^{f-m}) (\delta x_i \delta z_j)
\end{aligned} \tag{50}$$

The pressure equation and the transport equation in the rock matrix domain of the Full-3D model can be discretized following similar procedures as described above. The mass transfer functions in the Full-3D model are discretized as follows

$$\begin{aligned}
q_c^{f-m}{}_{i,j} = \sigma k^m{}_{i,j} & \left[\frac{k_{r,c,i,j}^f}{\mu_c} \delta_c^f{}_{i,j} + \frac{k_{r,c,i,j}^m}{\mu_c} (1 - \delta_c^f{}_{i,j}) \right] \left[p_b^f{}_{i,j} - p_b^m{}_{i,j} + (p^{cap})^f{}_{i,j} \right. \\
& \left. - (p^{cap})^m{}_{i,j} + \frac{\sigma_z}{\sigma} \Delta \rho g \left(\frac{s_{c,i,j}^f - s_{cr}^f}{1 - s_{cr}^f} - \frac{s_{c,i,j}^m - s_{cr}^m}{1 - s_{cr}^m} \right) l_z \right]
\end{aligned} \tag{51}$$

$$\begin{aligned}
q_b^{f-m}{}_{i,j} = \sigma k^m{}_{i,j} & \left[\frac{k_{r,b,i,j}^f}{\mu_b} \delta_b^f{}_{i,j} + \frac{k_{r,b,i,j}^m}{\mu_b} (1 - \delta_b^f{}_{i,j}) \right] \left[p_b^f{}_{i,j} - p_b^m{}_{i,j} \right. \\
& \left. + \frac{\sigma_z}{\sigma} \Delta \rho g \left(\frac{s_{b,i,j}^f - s_{br}^f}{1 - s_{br}^f} - \frac{s_{b,i,j}^m - s_{br}^m}{1 - s_{br}^m} \right) l_z \right]
\end{aligned} \tag{52}$$

Appendix B: Discretized Governing Equations for VE-3D

Model

In the fracture domain, the coarse-scale pressure equation is equation (39) and is repeated below

$$\begin{aligned} & (\Phi^f c_c^f + c_\phi^f) S_c^f \frac{\partial P_c^f}{\partial t} + (\Phi^f c_b^f + c_\phi^f) S_b^f \frac{\partial P_b^f}{\partial t} + \nabla_{\parallel} \cdot (\mathbf{U}_c^f + \mathbf{U}_b^f) \\ & = (\Psi^c)^f + (\Psi^b)^f - Q_c^{f-m} - Q_b^{f-m} \end{aligned} \quad (39)$$

Assuming no compressibility and only focusing on the (x,z) plane, the coarse-scale pressure equation becomes

$$\begin{aligned} & \frac{\partial}{\partial x} \left[-K^f \left(\frac{S_c^f}{1 - S_b^{res}} \frac{k_{r,c}^*}{\mu_c} \right) \frac{\partial P_c^f}{\partial x} \right] + \frac{\partial}{\partial x} \left\{ -K^f \left[\left(1 - \frac{S_c^f}{1 - S_b^{res}} \right) \frac{1}{\mu_b} \right] \frac{\partial P_b^f}{\partial x} \right\} \\ & = (\Psi^c)^f + (\Psi^b)^f - Q_c^{f-m} - Q_b^{f-m} \end{aligned} \quad (53)$$

All variables in equations (54) – (58) are from the fracture domain. Their superscripts f are omitted. Let P_{b_i} denotes P_b in the grid cell i . Because equation (53) is one-dimensional, there is no j in the equation. The coarse-scale pressure equation becomes

$$\begin{aligned}
& \frac{\left(\frac{2}{\frac{1}{K_i} + \frac{1}{K_{i+1}}}\right) \left(\frac{1}{\mu_b} - \frac{S_{c_i}}{\mu_b(1 - S_b^{res})} + \frac{k_{r,c_i}^*}{\mu_c} \frac{S_{c_i}}{1 - S_b^{res}}\right) \left(\frac{P_{b_{i+1}} - P_{b_i}}{\Delta x_i}\right)}{\delta x_i} \\
& - \frac{\left(\frac{2}{\frac{1}{K_i} + \frac{1}{K_{i-1}}}\right) \left(\frac{1}{\mu_b} - \frac{S_{c_i}}{\mu_b(1 - S_b^{res})} + \frac{k_{r,c_i}^*}{\mu_c} \frac{S_{c_i}}{1 - S_b^{res}}\right) \left(\frac{P_{b_i} - P_{b_{i-1}}}{\nabla x_i}\right)}{\delta x_i} \\
& + \frac{\left(\frac{2}{\frac{1}{K_i} + \frac{1}{K_{i+1}}}\right) \left(\frac{k_{r,c_i}^*}{\mu_c} \frac{S_{c_i}}{1 - S_b^{res}}\right) \left(\frac{P_{i+1}^{cap} - P_i^{cap}}{\Delta x_i}\right)}{\delta x_i} \\
& - \frac{\left(\frac{2}{\frac{1}{K_i} + \frac{1}{K_{i-1}}}\right) \left(\frac{k_{r,c_i}^*}{\mu_c} \frac{S_{c_i}}{1 - S_b^{res}}\right) \left(\frac{P_i^{cap} - P_{i-1}^{cap}}{\nabla x_i}\right)}{\delta x_i} \\
& = \Psi^c_i + \Psi^b_i - Q_c^{f-m}_i - Q_b^{f-m}_i
\end{aligned} \tag{54}$$

Multiply both sides of equation (54) by δx_i and expand

$$\begin{aligned}
& \left(\frac{2}{\frac{1}{K_i} + \frac{1}{K_{i+1}}} \right) \left(\frac{1}{\mu_b} - \frac{S_{c_i}}{\mu_b(1 - S_b^{res})} + \frac{k_{r,c_i}^*}{\mu_c} \frac{S_{c_i}}{1 - S_b^{res}} \right) \left(\frac{1}{\Delta x_i} \right) P_{b_{i+1}} \\
& - \left[\left(\frac{2}{\frac{1}{K_i} + \frac{1}{K_{i+1}}} \right) \left(\frac{1}{\mu_b} - \frac{S_{c_i}}{\mu_b(1 - S_b^{res})} + \frac{k_{r,c_i}^*}{\mu_c} \frac{S_{c_i}}{1 - S_b^{res}} \right) \left(\frac{1}{\Delta x_i} \right) \right. \\
& + \left. \left(\frac{2}{\frac{1}{K_i} + \frac{1}{K_{i-1}}} \right) \left(\frac{1}{\mu_b} - \frac{S_{c_i}}{\mu_b(1 - S_b^{res})} + \frac{k_{r,c_i}^*}{\mu_c} \frac{S_{c_i}}{1 - S_b^{res}} \right) \left(\frac{1}{\nabla x_i} \right) \right] P_{b_i} \\
& + \left(\frac{2}{\frac{1}{K_i} + \frac{1}{K_{i-1}}} \right) \left(\frac{1}{\mu_b} - \frac{S_{c_i}}{\mu_b(1 - S_b^{res})} + \frac{k_{r,c_i}^*}{\mu_c} \frac{S_{c_i}}{1 - S_b^{res}} \right) \left(\frac{1}{\nabla x_i} \right) P_{b_{i-1}} \\
& + \left(\frac{2}{\frac{1}{K_i} + \frac{1}{K_{i+1}}} \right) \left(\frac{k_{r,c_i}^*}{\mu_c} \frac{S_{c_i}}{1 - S_b^{res}} \right) \left(\frac{1}{\Delta x_i} \right) P_{i+1}^{cap} \\
& - \left[\left(\frac{2}{\frac{1}{K_i} + \frac{1}{K_{i+1}}} \right) \left(\frac{k_{r,c_i}^*}{\mu_c} \frac{S_{c_i}}{1 - S_b^{res}} \right) \left(\frac{1}{\Delta x_i} \right) \right. \\
& + \left. \left(\frac{2}{\frac{1}{K_i} + \frac{1}{K_{i-1}}} \right) \left(\frac{k_{r,c_i}^*}{\mu_c} \frac{S_{c_i}}{1 - S_b^{res}} \right) \left(\frac{1}{\nabla x_i} \right) \right] P_i^{cap} \\
& + \left(\frac{2}{\frac{1}{K_i} + \frac{1}{K_{i-1}}} \right) \left(\frac{k_{r,c_i}^*}{\mu_c} \frac{S_{c_i}}{1 - S_b^{res}} \right) \left(\frac{1}{\nabla x_i} \right) P_{i-1}^{cap} \\
& = - \left(\Psi_{c_i}^c + \Psi_{b_i}^b - Q_c^{f-m} - Q_b^{f-m} \right) \delta x_i
\end{aligned} \tag{55}$$

The coarse-scale transport equation in the fracture domain is equation (40) and is repeated below

$$\Phi^f \frac{\partial S_c^f}{\partial t} + \left(\Phi^f c_c^f + c_\phi^f \right) S_c^f \frac{\partial P_c^f}{\partial t} + \nabla_{\parallel} \cdot \mathbf{U}_c^f = (\Psi^c)^f - Q_c^{f-m} \tag{40}$$

Assuming no compressibility and only focusing on the (x,z) plane, the coarse-scale transport equation becomes (56) and is discretized into equation (57)

$$\Phi \frac{\partial S_c}{\partial t} + \frac{\partial}{\partial x} \left[-K \left(\frac{S_c}{1 - S_b^{res}} \frac{k_{r,c}^*}{\mu_c} \right) \frac{\partial P_c}{\partial x} \right] = \Psi^c - Q_c^{f-m} \quad (56)$$

$$\Phi \frac{\partial S_c}{\partial t} \delta x_i + \frac{K \left(\frac{S_{c,i}}{1 - S_b^{res}} \frac{k_{r,c,i}^*}{\mu_c} \right) \frac{\partial P_c}{\partial x} \Big|_{i+\frac{1}{2}} - K \left(\frac{S_{c,i}}{1 - S_b^{res}} \frac{k_{r,c,i}^*}{\mu_c} \right) \frac{\partial P_c}{\partial x} \Big|_{i-\frac{1}{2}}}{\delta x_i} = -(\Psi^c_i - Q_c^{f-m}_i) \quad (57)$$

Multiply both sides of the equation by δx_i . Equation (57) can be expressed as

$$\begin{aligned} & \Phi \frac{S_{c,i}^{n+1} - S_{c,i}^n}{\Delta t} \delta x_i + \left(\frac{2}{\frac{1}{K_i} + \frac{1}{K_{i+1}}} \right) \left(\frac{k_{r,c,i}^*}{\mu_c} \frac{S_{c,i}}{1 - S_b^{res}} \right) \left(\frac{1}{\Delta x_i} \right) P_{c,i+1} \\ & - \left[\left(\frac{2}{\frac{1}{K_i} + \frac{1}{K_{i+1}}} \right) \left(\frac{k_{r,c,i}^*}{\mu_c} \frac{S_{c,i}}{1 - S_b^{res}} \right) \left(\frac{1}{\Delta x_i} \right) \right. \\ & \left. + \left(\frac{2}{\frac{1}{K_i} + \frac{1}{K_{i-1}}} \right) \left(\frac{k_{r,c,i}^*}{\mu_c} \frac{S_{c,i}}{1 - S_b^{res}} \right) \left(\frac{1}{\nabla x_i} \right) \right] P_{c,i} \\ & + \left(\frac{2}{\frac{1}{K_i} + \frac{1}{K_{i-1}}} \right) \left(\frac{k_{r,c,i}^*}{\mu_c} \frac{S_{c,i}}{1 - S_b^{res}} \right) \left(\frac{1}{\nabla x_i} \right) P_{c,i-1} = -(\Psi^c_i - Q_c^{f-m}_i) \delta x_i \end{aligned} \quad (58)$$

The pressure equation and the transport equation in the rock matrix domain of the VE-3D model can be discretized following similar procedures as described in Appendix A. The coarse-scale mass transfer functions in the VE-3D model are discretized as follows

$$\text{Let } a_c = \sigma k^m \left[\frac{k_{r,c}^f}{\mu_c} \delta_c^f + \frac{k_{r,c}^m}{\mu_c} (1 - \delta_c^f) \right] \text{ and } b_c = \frac{\sigma_z}{\sigma} \Delta \rho g \left(\frac{s_c^f - s_{cr}^f}{1 - s_{cr}^f} - \frac{s_c^m - s_{cr}^f}{1 - s_{cr}^f} \right) l_z$$

$$\text{Let } a_b = \sigma k^m \left[\frac{k_{r,b}^f}{\mu_b} \delta_b^f + \frac{k_{r,b}^m}{\mu_b} (1 - \delta_b^f) \right] \text{ and } b_b = \frac{\sigma_z}{\sigma} \Delta \rho g \left(\frac{s_c^f - s_{cr}^f}{1 - s_{cr}^f} - \frac{s_c^m - s_{cr}^f}{1 - s_{cr}^f} \right) l_z$$

$$\begin{aligned}
Q_c^{f-m}{}_i &= \sum_{j=1}^{Nz} a_{c,i,j} \cdot P_{b,i}^f \cdot \delta z + \sum_{j=1}^{Nz} a_{c,i,j} \cdot P_i^{cap,f} \cdot \delta z - \sum_{j=1}^{Nz} \rho_c g \cdot a_{c,i,j} \cdot z_{i,j} \cdot \delta z \\
&\quad - \sum_{j=1}^{Nz} a_{c,i,j} \cdot p_{b,i,j}^m \cdot \delta z - \sum_{j=1}^{Nz} a_{c,i,j} \cdot p_{i,j}^{cap,m} \cdot \delta z + \sum_{j=1}^{Nz} a_{c,i,j} \cdot b_{c,i,j} \cdot \delta z \\
&= \left(\sum_{j=1}^{Nz} a_{c,i,j} \right) \cdot \delta z \cdot P_{b,i}^f + \left(\sum_{j=1}^{Nz} a_{c,i,j} \right) \cdot \delta z \cdot P_i^{cap,f} - \left(\sum_{j=1}^{Nz} a_{c,i,j} \cdot z_{i,j} \right) \cdot \delta z \cdot \rho_c g \\
&\quad - \left(\sum_{j=1}^{Nz} a_{c,i,j} \cdot p_{b,i,j}^m \right) \cdot \delta z - \left(\sum_{j=1}^{Nz} a_{c,i,j} \cdot p_{i,j}^{cap,m} \right) \cdot \delta z \\
&\quad + \left(\sum_{j=1}^{Nz} a_{c,i,j} \cdot b_{c,i,j} \right) \cdot \delta z
\end{aligned} \tag{59}$$

$$\begin{aligned}
Q_b^{f-m}{}_i &= \sum_{j=1}^{Nz} a_{b,i,j} \cdot P_{b,i}^f \cdot \delta z - \sum_{j=1}^{Nz} \rho_b g \cdot a_{b,i,j} \cdot z_{i,j} \cdot \delta z - \sum_{j=1}^{Nz} a_{b,i,j} \cdot p_{b,i,j}^m \cdot \delta z \\
&\quad - \sum_{j=1}^{Nz} a_{b,i,j} \cdot b_{b,i,j} \cdot \delta z \\
&= \left(\sum_{j=1}^{Nz} a_{b,i,j} \right) \cdot \delta z \cdot P_{b,i}^f - \left(\sum_{j=1}^{Nz} a_{b,i,j} \cdot z_{i,j} \right) \cdot \delta z \cdot \rho_b g \\
&\quad - \left(\sum_{j=1}^{Nz} a_{b,i,j} \cdot p_{b,i,j}^m \right) \cdot \delta z - \left(\sum_{j=1}^{Nz} a_{b,i,j} \cdot b_{b,i,j} \right) \cdot \delta z
\end{aligned} \tag{60}$$